

Casualties of border changes

SUMMARY

We investigate the economic effects of the Russia–Ukraine conflict – following the 2014 annexation of Crimea – on Russian border regions. While southern regions gained market access to Crimea, northern regions lost market access to Ukraine. Using nighttime lights data and geo-referenced plant-level data, we find that regions with deteriorating market access saw 43% less growth in lights – translating into 6–12% lower growth in GDP depending on the assumed lights–GDP elasticity – and a 35% increase in the exit probability for manufacturing plants after 2014. Exploiting variations in closed local border crossings in the northern regions, we find that these effects may be partly driven by less cross-border labour flows.

JEL codes: F51, F15, R11, R12

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Casualties of border changes: evidence from nighttime lights and plant exit

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1. INTRODUCTION

We exploit the Russia–Ukraine conflict – following the 2014 annexation of Crimea – as a natural experiment to provide evidence for the effects of conflict-induced market access changes on the economic performance of border regions. Using nighttime lights satellite data and geo-referenced manufacturing plant data, we find that the Russian regions along the northern land border with Ukraine – which saw their market access deteriorate due to border restrictions – performed worse than Russian regions in the south that benefitted from an increase in market access due to the annexation of the close-by Crimea. The areas most exposed to economic activity in Ukraine before 2014 and to border restrictions after 2014 saw their luminosity decrease by about 10%, whereas the areas the least exposed to economic activity in Ukraine before 2014 and that benefitted for an increase in market access after 2014 saw their luminosity increase

* I thank the editor Mathias Thoenig, two anonymous reviewers, my discussant Federico Rossi, as well as Julien Martin, Florian Mayneris and Marlon Seror for helpful discussions and suggestions. I am also grateful to seminar and conference participants at the 77th CEPR Policy panel at the Sveriges Riksbank, UQAM, Michigan State University, the University of Oregon and the University of Trento for useful comments and suggestions. Financial support from the SSHRC Insight Grants program ('Cities in motion', grant #435-2016-1246; and 'Neighborhood change and businesses', grant #435-2021-1017) is gratefully acknowledged.

The Managing Editor in charge of this paper was Mathias Thoenig, Guest Editor of the Special Issue on Geoeconomics.

by about 33%. This 43% difference in the changes in lights translates into a 6–12% difference in per-capita GDP growth, depending on the estimates of the GDP-nighttime lights elasticity. Using manufacturing plant exit as an alternative measure for economic outcomes, we find that the exit probability of the least exposed plants increased by about 15% after 2014, whereas that of the most exposed plants increased by about 50%. Hence, the most exposed plants had a 35% larger increase in their exit probability in the wake of the annexation.

Understanding the *causes and consequences* of territorial conflict and border changes is important. It is well understood now that territorial conflict is more likely and prone to escalation when: (1) there are contested boundaries, that is, borders do not play their institutional role of creating a predictable environment (e.g., [Hensel, 2000](#); [Simmons, 2005](#)); (2) previously integrated territories have become independent or have been split, creating new borders (e.g., [Resnick, 2012](#); [Michalopoulos and Papaioannou, 2016](#); [Cederman et al., 2022](#)); and (3) there are significant cross-border ethnic ties and a shared history (e.g., [Resnick, 2012](#)). Conflict is also more likely between contiguous neighbours since political events closer to home are perceived as more threatening and because military force can be projected more easily close to home ([Hensel, 2000](#)). We explain in Section 2 – when reviewing the geopolitical context – that the Russia–Ukraine conflict seems driven by all of these usual factors. Our paper, therefore, has not much to add to the literature on the *causes* of territorial conflict.

We instead contribute to the growing literature on the *economic consequences of conflict*. We are particularly interested in the effects of changes in local market access – driven by conflict – on economic outcomes, irrespective of direct conflict-related destruction. To this end, we look at the Russian border regions (excluding Crimea) – which were not a direct scene of conflict after 2014 – to isolate the effects of changes in market access on economic outcomes. Our paper complements recent work by [Zhukov \(2016\)](#), who documents the conflict-related costs on the Ukrainian side for the Donbass (Donetsk and Luhansk regions), and [Korovkin et al. \(2023a,b\)](#), who analyse and quantify the economic consequences of the conflict for the Ukrainian regions outside the Donbass that were not directly exposed to combat.¹ We look to the other side of the border and show that the conflict – which in 2014 took place entirely on Ukrainian territory – had differential effects on the Russian border regions.

One challenge in identifying the causal effects of changes in market access after 2014 is that the annexation triggered a sharp international response: wide-ranging trade sanctions were imposed on Russia – which responded with its own import restrictions – and the ruble devalued substantially. To tackle this issue, we exploit north–south differences in

1 The conflict largely unfolded in the Donbass and in Crimea. [Zhukov \(2016\)](#) and [Kochnev \(2019\)](#) document the conflict-related costs on the Ukrainian side for the Donetsk and Luhansk regions. For more results on the economic effects of conflict on the geography of economic activity and firm-level outcomes in conflict regions, see, for example, [Abadie and Gardeazabal \(2003\)](#), [Guidolin and La Ferrara \(2007\)](#), [Camacho and Rodriguez \(2013\)](#), [Collier and Duponchel \(2013\)](#), [Berman and Couttenier \(2015\)](#) and [Harari and Ferrara \(2018\)](#).

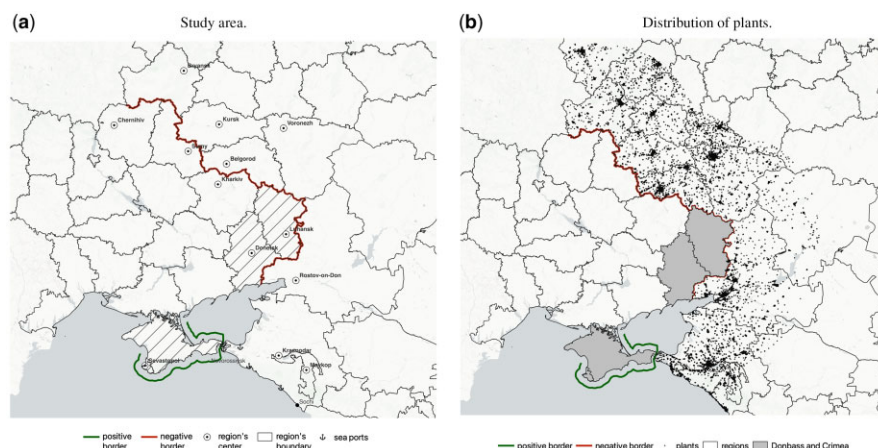


Figure 1. Study area and distribution of Russian manufacturing plants along the Ukrainian border

Notes: Panel (a) shows the major cities along the border on both sides and highlights merchant sea ports. Panel (b) depicts the distribution of manufacturing plants in Russia from the Ruslana and Interfax SPARK databases within 300 km from the border. We plot the distribution of all plants active at some point between 2006 and 2018. Both panels show the border segments. The ‘positive border segment’ (in green) – where market access likely improved – is in the south close to Crimea, whereas the ‘negative border segment’ (in red) – where market access likely deteriorated – is in the north.

market access changes across regions (see Figure 1a for our study region along the Russia–Ukraine border). While international sanctions likely affected regions in similar ways, market access to Ukraine deteriorated more in the north – where closed border crossings and tighter border controls limited the movements of goods and people across the border – and improved in the south – where the border with Crimea disappeared and market access to the previously Ukrainian region became de facto unfettered. We construct several measures of initial exposure to Ukraine and changes in market access and show how they affect economic outcomes as measured by nighttime lights and plant exit.

Our findings confirm the importance of pre-conflict exposure and changes in market access on economic outcomes. Furthermore, the effects differ substantially across regions, depending on the regions’ pre-conflict exposure to Ukraine and their exposure to changes in border frictions. These results are robust and hold for a broad range of measures: border regions in Russia more exposed to negative changes in market access – namely the regions in the north – performed less well: they saw less growth in lights and more plant exit than less exposed border regions – namely the regions in the south.²

2 Ideally, we would measure GDP at a small geographic scale and use it as our dependent variable. However, these data obviously do not exist. We thus use nighttime lights which, as previous work has shown, are a good proxy for GDP and which have been widely used in settings such as ours where sub-national quality data are not available (e.g., Chen and Nordhaus, 2011; Henderson et al., 2012; Michalopoulos and Papaioannou, 2012; Hodler and Raschky, 2014; Pinkovskiy and Sala I Martin, 2016; Bruederle and Hodler, 2018). As a robustness check, we spatially disaggregate regional GDP

To shed additional light on the economic mechanisms that may drive the results, we look at more granular geographic and temporal data and examine quarterly changes *within* the four northern Russian regions bordering Ukraine (Voronezh, Bryansk, Kursk and Belgorod; see [Figure 1a](#)). We leverage new data we assembled on the closings of specific local border crossings after March 2015, which affected the movement of people but not of commercial merchandise. We find that locations that experienced a larger increase in their distance to the closest open border crossing saw less growth in nighttime lights. We do, however, not find substantial differences in plant exit rates. Our finding for nighttime lights suggests that very distance-sensitive local economic interactions – such as local cross-border labour movements or cross-border shopping – may partly explain the localized nature and magnitudes of our effects.

The remainder of the paper is organized as follows. We first provide an overview of the related literature. Section 2 then summarizes the geo-political context. Section 3 explains our data, the construction of our variables and provides preliminary descriptive evidence. Section 4 shows our estimation results. We provide a detailed discussion on possible confounders in Section 5. Section 6 zooms in on the northern border regions and shows the effects of the closing of local border crossings. Last, Section 7 discusses the policy relevance of our contribution and concludes. We relegate additional details to an extensive set of [Supplementary Appendices](#).

1.1. Related literature

Our paper is related to several strands of literature. First, it is related to two recent papers that investigate the effects of localized conflict in the Donbass on the economic performance of non-combat areas in Ukraine. Korovkin and Makarin (2023) study how trade between Ukrainian regions and Russia changed after 2014 based on the regions' ethnic shares of Russians. They find that trade with Russia fell substantially more for regions having a smaller share of ethnic Russians and that firms modified their types of contracts and timing of payments. These results are in line with a trust-based explanation of trade (e.g., [Rohner et al., 2013](#)) and show how the effects of conflict spill over onto non-combat areas. In a follow-up paper, [Korovkin et al. \(2023a,b, p. 5\)](#) study how the 'economic cost of conflict extends beyond the direct conflict regions through the disruption and reorganization of production networks'. They document how firms reorganized their buyer–supplier links in the wake of 2014 to reduce their reliance on the conflict area in the Donbass. Using a structural model, they estimate a 17% reduction

using cell-level lights as weights. Both results are very similar. We show in [Appendix C](#) (see Figures C1–C3) that the harmonized nighttime lights satellite data we use correlate well with regional per capita GDP, as well as with municipal employment, wages and the number of manufacturing plants.

in welfare for the average region relative to a region that would not have any buyer–supplier links with the Donbass before 2014.

Our paper is further related to a recent but growing literature that investigates the economic consequences – on nighttime lights and other measures such as schooling, population growth or agricultural productivity – of conflict-induced changes in market access. Most closely related to our research are recent contributions that focus on the legacy of landmines or, more generally, unexploded ordnance (UXO). [Chiovelli *et al.* \(2018\)](#) quantify the economic consequences of landmine clearance in Mozambique using nighttime lights and a market access perspective. They find that ‘aggregate light density in 2015 would have been roughly 67% lower [in the absence of clearance]. Given a lights-GDP elasticity of around 0.25 ([Henderson *et al.*, 2012](#)), this implies an output boost of around 16% from clearance’. ([Chiovelli *et al.*, 2018](#), p. 3). They also document substantial heterogeneity: clearing areas located close to the transportation network or areas that are local markets maps into larger gains, which suggests a role for quite local changes in market access. In a similar vein, [Prem *et al.* \(2023\)](#) quantify the effects of demining in Colombia. They find that, following humanitarian demining campaigns, nightlights increase by 12.4% on average over the next 3 years. The effects are again heterogeneous, with areas better connected via roads benefitting more. They conclude that ‘absent any humanitarian effort, nighttime luminosity would have been 17 percent lower between 2013 and 2019 [...] using the GDP to nighttime lights elasticity of 0.3 [...] this implies that, over that period, Colombia grew an additional 5.1 percent thanks to humanitarian demining’ ([Prem *et al.*, 2023](#), p. 23).

Both aforementioned papers estimate the costs of reduced market access due to landmines. Another legacy of war – with similar effects – is UXO that still litters areas in countries such as Laos and Cambodia. [Riaño and Caicedo \(2024\)](#) and [Lin \(2022\)](#) investigate the negative consequences of UXO in those two countries. Using detailed information on US bombing campaigns, the former paper finds that ‘a one standard deviation increase in bombs leads to a 33% decrease in nightlights with respect to their mean [...] an effect that corresponds to a 9.3% fall in GDP per capita, a sizable decrease’ ([Riaño and Caicedo, 2024](#), p. 3). They further find that areas that were bombed more heavily are less densely populated today and have lower literacy rates than areas that were less bombed. [Lin \(2022\)](#) provides a similar analysis for Cambodia, exploiting variation in UXO due to different failure rates of bombs linked to soil conditions. She finds that ‘shifting from no bombing to the sample mean, the rice surplus also drops 18% [...] This represents a 44% decline in market sales. [...] farmers produce 50% less rice and collect 60% less income than their counterparts on non-bombed, similarly-fertile land’ ([Lin, 2022](#), pp. 232–235). It is worth emphasizing that, contrary to other papers that have looked at Japan or Germany (e.g., [Davis and Weinstein, 2002](#); [Brakman *et al.*, 2004](#)), both aforementioned papers find a persistent negative long-run effect of the bombing campaigns. This ‘legacy of war’ is due to the presence of UXO. This points to one mechanism by which the deleterious effects of conflict can extend for

a long time and reduce market access and economic outcomes long after the end of the conflict *per se*.

Last, our paper generally relates to a large literature on the effects of changes in economic integration, whatever the reasons for those changes. One strand of that literature investigates the aggregate effects of conflict on trade between countries. As expected, conflict reduces trade and generates substantial economic costs.³ Other contributions focus on the economic effects of borders on the location and size of economic activity (e.g., [Hanson, 1996](#); [Redding and Sturm, 2008](#); [Brülhart *et al.*, 2012](#); [Ahlfeldt *et al.*, 2015](#); [Brülhart *et al.*, 2018](#)). That literature shows that border regions that experience a negative shock to market access see less population and wage growth than unaffected regions. More recently, many contributions have leveraged spatially and economically more disaggregated data – namely nighttime lights satellite data and individual-level or firm-level data – to study the effects of market access on regional outcomes. In a recent paper, [Emran *et al.* \(2019\)](#) investigate how the civil war in Côte d'Ivoire cut off Abidjan as a harbour for exports from the neighbouring Mali and Burkina Faso. They document that the conflict-induced disruption of market access implied that the areas not relying on the route to Abidjan had a 36% higher growth in nightlights during the war period, which corresponds to a 10.8% higher growth in GDP. They find that these effects are persistent. In other closely related studies, [Lee \(2018\)](#) uses nighttime lights to investigate the unequal regional effects of sanctions in North Korea. He finds that 'difference in nighttime lights between the capital Pyongyang and the rest of the country increases by 1.9% with an additional sanctions event. This translates to about 0.6% in terms of GDP' ([Lee, 2018](#), p. 35). Last, [Eberhard-Ruiz and Moradi \(2019\)](#) look at a regional trade agreement in Africa (EAC) and find that nighttime lights 'were 13% brighter than they would have been without the launch of the EAC' ([Eberhard-Ruiz and Moradi, 2019](#), p. 264), which translates into a 5% difference in GDP. They attribute the localized effects close to the border to the positive effects of market access on economic activity that is very sensitive to distance frictions such as small-scale local cross-border trade.

2. GEOPOLITICAL CONTEXT AND SETTING

Russia and Ukraine – separated since 1991 – always maintained a high level of integration, especially in the (south)eastern part where cross-border industrial and ethnic ties were strong. Although a new border *de facto* appeared in 1991, following the collapse

3 [Simmons \(2005, p. 842\)](#) finds that 'border disputes have led to serious economic opportunity costs, even in cases where trade partners have never exchanged an explicit military threat'. The presence of a territorial dispute between country pairs is associated with a 28% decline in the value of their bilateral trade in the short run. [Martin *et al.* \(2008\)](#) show that increasing multilateral trade openness may explain why trade is not as strongly associated with less conflict as one would expect: less trade with neighbours in an increasingly globalized world reduces the opportunity cost of conflict with neighbours.

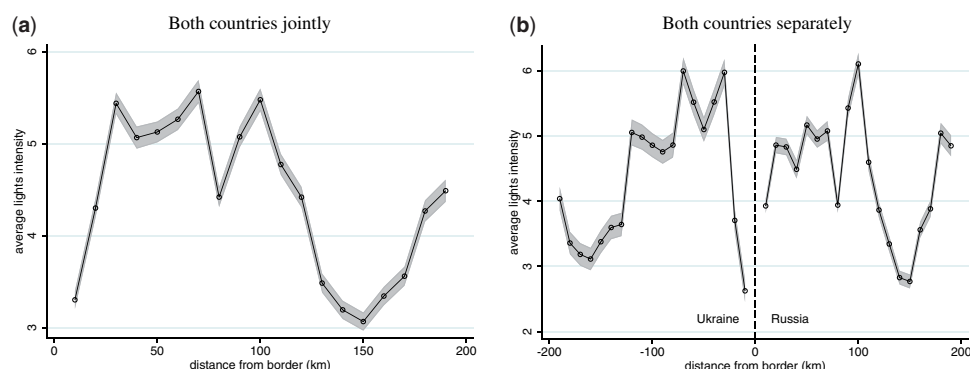


Figure 2. Non-parametric nighttime lights distance gradients

Notes: Non-parametric estimates of nighttime lights distance gradients using distance-bin dummies between 0 and 200 km for 10 km bins. The 95% confidence bands are shaded in gray. Nighttime lights are measured for 1×1 km cells, excluding water-masked cells. Panel (a) depicts the distance gradient for both countries jointly, while panel (b) depicts it for each country separately.

of the Soviet Union, that border remained largely symbolic and was not enshrined in a formal treaty. Starting in the early 2000s – in the wake of the Orange Revolution and the arrival of a more pro-western president – Ukraine progressively sought to distance itself from the former Soviet bloc by turning increasingly to the West. The ensuing tensions – one of the debated questions was already the status of the Crimean peninsula – led to the ratification of a formal border treaty by the parliaments of both countries in April 2004.⁴

Yet, even following the treaty, the degree of economic integration between border regions in the two countries remained strong. This can be seen using nighttime lights data. Figure 2 depicts non-parametric gradient estimates of average 1992–2013 lights intensities as a function of distance to the border.⁵ Up to about 30 km the gradient increases; it then has a large plateau on both sides of the border and starts to decrease again beyond 100 km. The average nighttime lights intensity in the 0–100 km distance band from the border is 25.15% *brighter* on the Russian side and 18.47% *brighter* on the Ukrainian side as compared to that in the 100–200 km distance band. This differs from the findings by Brühlhart *et al.* (2019), who document that from 1995 to 2013 nighttime lights in the world were on average 37% *dimmer* at a land border between countries than 200 km away from that border. This shows that the Russia–Ukraine border was

4 The Crimean peninsula was transferred under the Soviet leader Nikita Khrushchev to Ukraine by a decree of the RSFSR Council of Ministers on 5 February 1954, in what is believed to be a move to strengthen the Russian influence in Ukraine. After Ukrainian independence in 1991, Crimea was granted the status of autonomous republic. Its major city, Sevastopol, also enjoyed a special status. Crimea has always played a special role for Russia due to its strategic location on the Black Sea. It hosts most of the Russian Black Sea fleet and allows to control access to the Sea of Azov via the Strait of Kerch.

5 Following Brühlhart *et al.* (2022) and, more generally, a large literature on gradient estimates in spatial economics, we depict non-parametric estimates of the distance gradient using distance-bin dummies.

atypical, with substantial population and economic activity located close-by in the strongly integrated border regions.

Despite the high level of integration, relations between Russia and Ukraine started to deteriorate after 2004 as delicate questions such as stronger border demarcations, a European orientation, NATO membership, the role of Sevastopol as a base for the Black Sea fleet, and gas transit through Ukraine made it on the agenda. Russia increasingly feared its weaker ties with Ukraine. During 2010–2013, the relationship stabilized as agreements concerning the Black Sea naval bases and preferential gas supply were reached. Ukraine also signed the CIS Free Trade Agreement that brought it back more closely into the Russian orbit. Yet, the question of a tighter integration with the European Union (EU) remained.

In early 2012, the EU and Ukraine initialized the EU Association agreement, which was approved by Ukraine in September 2013. This prompted strong reactions from Moscow, which questioned the recognition of the bilateral border treaty.⁶ The Ukrainian government subsequently decided to suspend the signing of the Association Agreement in favour of closer ties to Russia and the Eurasian Economic Union. This sparked a wave of civil unrest – the ‘Euromaidan’ – which led to the ousting of the incumbent president.

In late February 2014, following the unrest, pro-Russian demonstrations were held in Sevastopol. A few days later, during the referendum of 16 March 2014, a majority of respondents expressed their desire for Crimea – populated by a majority of ethnic Russians – to join Russia. Although Ukraine and the international community at large condemned the referendum, Crimea de facto joined Russia via the accession agreement signed in Moscow on 18 March 2014 and two new federal subjects – the Republic of Crimea and the federal city of Sevastopol – were created. In the wake of the annexation, two eastern regions of Ukraine – Donetsk and Luhansk, collectively known as Donbass – carried out their own local referenda. As a result, two republics were self-proclaimed on April 8 and 27 April 2014, respectively. An armed conflict with Ukrainian forces in the Donbass followed these declarations of independence.

Our objective is to better understand and measure the potential economic costs of the conflict following the annexation of Crimea for the border regions in Russia. This conflict provides a good laboratory to study the economic consequences of changes in international borders for two reasons. First, as discussed above, Russia and Ukraine were highly integrated – especially in the border regions. Hence, any changes in borders there are likely to have important economic effects. A first look at the changes in lights intensities before and after 2014 (see [Figure 3b](#) below) reveals the potential effect of the conflict on economic activity and suggests a decrease in border integration: between 2014 and 2018, the 0–100 km ‘lights premium’ drops to 9.85% on the Russian side and

6 Sergei Glazyev, adviser to President Vladimir Putin at that time, ‘suggested that if Ukraine signed the agreement, Russia would consider the bilateral treaty that delineates the countries’ borders to be void’ (The Guardian, 22 September 2013).

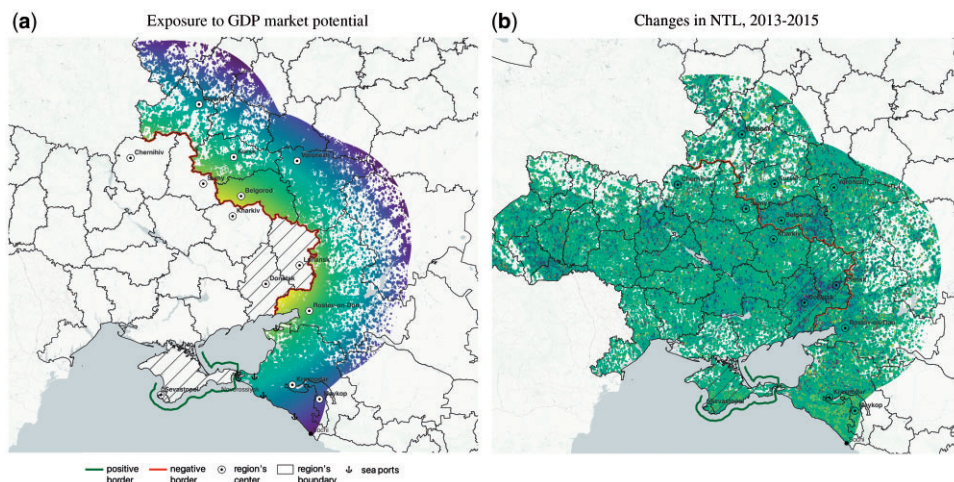


Figure 3. Exposure to market potential and changes in lights

Notes: Panel (a) shows the 5 years average 2009–13 pre-conflict exposure measure of cells in Russian border regions to regional GDP in Ukraine weighted by nighttime lights. It is computed as in Equation (A.2) with $\alpha = 0.01$. Panel (b) shows 2013–2015 changes in $\ln(\omega_{it} + 1)$. Lighter colours (towards yellow) represent higher market potential and larger NTL growth, whereas darker colours (towards blue) represent lower market potential and lower NTL growth. We plot changes in terms of standard deviations from the mean change in each country for each 1×1 kilometre grid cell. We suppress all water-masked cells and ‘empty’ cells that never show any lights emissions.

9.86% on the Ukrainian side. In other words, regions closer to the border suffered more in the wake of the conflict than regions further away, at least as revealed by night-time lights.

Second, while border regions seemed to suffer from the conflict, the averages we report above mask substantial heterogeneity. Following the annexation of Crimea the border between Russia and Ukraine de facto disappeared in the south, whereas it was subject to substantially tighter restrictions from the Ukrainian authorities in the north.⁷ We can thus investigate finely how differential changes in market access affect economic activity in the same period. We conjecture that the regions close to the southern sea border with Crimea – which are exposed to *positive border changes* following the annexation – benefit from increased market access and less frictions to the movement of goods and people; whereas the regions close to the northern land border with Ukraine – which were more exposed to *negative border changes* due to tighter controls – suffered from decreased market access and more frictions to the movement of goods and people. Thus, if market access matters, economic outcomes should improve in the south relative to those in the north.

⁷ What happened to the border along the Donbass region is less clear, since it was no longer fully under the control of the Ukrainian government and hence movements across that border were hard to monitor.

3. DATA, VARIABLES AND FIRST DESCRIPTIVES

We first provide information on our main data sources, the construction of our measures of economic outcomes, and the construction of our measures of exposure to Ukraine and to border changes.

3.1. Data and variables

Our key dependent variables are based on cell-level nighttime lights and plant-level exit information. For nighttime lights, we use the yearly series of Harmonized NTL (henceforth, HNTL) developed by [Li et al. \(2020\)](#) and based on DMSP and VIIRS.⁸ The HNTL data cover the period 1992–2018 and provide a consistent time series at a resolution of 1×1 km grid cells. We extract all cells for the European part of Russia (3,886,810 cells) and for Ukraine (811,399 cells).⁹ In what follows, we refer to the European part of Russia as Russia for short.

Formally, our measure for cell i in year t is defined as follows:

$$\omega_{it} = \text{NTL}_{it} \quad (1)$$

where NTL_{it} denotes cell i 's raw NTL luminosity, a digital number between 0 (no lights) and 63 (maximum top-coded lights). In [Appendix A.1](#), we also explain how we can use NTL to apportion regional GDP to cells based on luminosity. This allows us to partly control for regional differences due to, for example, specialization in high versus low value-added activity or more urban versus more rural regions. Our qualitative results are identical and the key magnitudes are very similar to those we obtain based on raw NTL. We thus consider those results as robustness checks and relegate them to the Appendix.

For plant-level exit, we use proprietary data for Russian manufacturing plants from Bureau van Dijk's Ruslana and Interfax's SPARK. We extract all plants that were active at some point between 2006 and 2018. Our data provide detailed information on active plants and those that enter or exit. For most plants, we know the exact date of entry and exit – as recorded in the Unified State Register of Legal Entities – as well as their de facto address, main national industry classification (OKVED), age, and whether it belongs to a multi-unit firm.¹⁰ We create a variable ω_{pt} taking value 1 if plant p exits in year t and value 0 otherwise. We drop all plant–year observations for the years the

8 Additional details on the NTL data are relegated to [Appendix A.1](#). We will indicate the rare cases where we rely on other satellite data such as the quarterly VIIRS series in Section 6.

9 The European part of Russia is bounded to the east by the Ural (see [Figure 1](#) in [Aleksandrova et al., 2020](#) for a map). Cell counts exclude water-masked cells and unlit cells that report zeros in all years.

10 The remaining variables in the dataset – size and various legal and financial indicators – are too sparse and non-representative to be useful. Only listed firms are required to provide that information and even then there are too many missing observations. Additional details on the data are given in [Appendix A.2](#).

plant did not exist (i.e., all years prior to entry and all years past exit).¹¹ We define exit based on the status updates from the Unified State Register of Legal Entities. Close to 90% of what we call ‘exit’ is administratively related to: (1) no more banking and fiscal activity of the plant in the last 90 days, meaning it has de facto ceased commercial operations or (2) a formal liquidation or bankruptcy procedure involving the plant. Since we focus on yearly exit, our measure captures well the timing of exit. We keep all plants in the European part of Russia and drop the small number we cannot geocode precisely or for which we cannot reconstruct industry codes. After basic data cleaning, we have 672,158 geo-referenced plants that are representative of Russian manufacturing. Figure 1b illustrates the distribution of plants along the border with Ukraine.

We now turn to the construction of our explanatory variables. To explore how economic outcomes in Russian border regions changed after 2014, we regress our outcomes on variables capturing either pre-conflict exposure to economic activity in Ukraine or exposure to changes in border frictions. Our measures can be divided into *absolute* exposure measures – which do not pay attention to the position of cells or plants with respect to the northern and southern parts of the border – and *relative* exposure measures – which do pay attention to the north–south dimension. We construct eight different measures.

Starting with our three absolute measures, we first create a dummy that takes value 1 for all plants or cells less than 150 km from the border, and 0 otherwise.¹² Second, we compute for each cell i and plant p two measures of market potential that capture their exposure to economic activity in Ukraine and in Russia. To alleviate notation, we provide expressions for cells only, those for plants being identical when replacing i with p . We define for each cell i in Russia its exposure to economic activity in Ukraine as follows:

$$\text{LMP}_{it}^{\text{UKR}} = \sum_{j \in \text{UKR}} \omega_{ji} e^{-\alpha d_{ij}}. \quad (2)$$

where ω_{ji} is given by Equation (1). Expression (2) is a (negative exponential) distance-weighted measure of cell i ’s exposure to nighttime lights in Ukraine. We can further compute Equation (2) by summing across cells in Russia (instead of Ukraine), which provides a measure of the cell’s or plant’s exposure to economic activity in Russia. We use the latter as a control in our subsequent regressions.¹³ We consider pre-treatment

11 We keep the few plants that enter and exit in the same year. Dropping them does not change our results. Since plants report the precise date of exit from the Unified State Register of Legal Entities, including day and month, we can further disaggregate the data to the quarterly level, which we use in Section 6.

12 Our results are robust to using alternative distance bands of 50 or 100 km.

13 We also combine information on NTL luminosity and regional GDP for Ukrainian and Russian regions to compute a GDP-based measure of market potential (see Appendix A.1 for details). We will use both lights- and GDP-based measures in our regressions. Expression (2) and (A.2) in the Appendix are not theory-based and require a value for α . We choose $\alpha = 0.01$ (we also checked our results for $\alpha = 0.05$) in our preferred specification and consider only cells in Russia or in Ukraine that are less than 500 kilometers from the border. These choices are made to reduce the computational

exposure based on market potential using the 5-year pre-conflict average exposure (2009–13) as follows:

$$\ln(\exp_i^{\text{NTL}}) = \ln\left(\frac{1}{5} \sum_{t=2009}^{2013} \text{LMP}_{it}^{\text{UKR}}\right) \quad (3)$$

The measure (3) captures whether cell i was strongly exposed to economic activity in Ukraine before the beginning of the conflict in 2014. We expect that plants or cells more exposed in the pre-treatment period suffered more than less exposed ones post-treatment. Figure 3a shows the distribution of market potential in the Russian border regions before the 2014 conflict.

Distance bands and market potential measures, as defined above, capture absolute exposure to Ukraine but do not treat the border in the north and in the south differently. We thus construct five relative measures that aim at capturing how cells or plants are exposed differently to the northern and southern parts of the border.

First, we consider distance bands that are specific to the borders in the north and in the south. For each cell and plant, we create two dummies, one for being close to each of the two borders. Second, we construct the ratio of the great circle (GC) distance from the positive border segment in the south to the negative border segment in the north:

$$\ln(\exp_i^{\text{GC}}) = \ln\left(\frac{\min_{j \in \mathcal{V}_P} d_{ij}^{\text{GC}}}{\min_{k \in \mathcal{V}_N} d_{ik}^{\text{GC}}}\right), \quad (4)$$

where $\mathcal{V}_P$ and $\mathcal{V}_N$ are the sets of vertices for the positive and negative border segments, respectively. Expression (4) is the ratio of the minimum distance to the border positively affected by the annexation of Crimea (i.e., the southern part of the border that experienced an increase in market access) and the minimum distance to the border negatively affected by the annexation of Crimea (i.e., the northern part of the border that experienced a decrease in market access due to tighter border restrictions).

Third, since crow-fly distance may not be a good approximation of travel distance, especially short ones, we compute the relative network distance (ND) on the main road system from each plant p to the positive and the negative border¹⁴:

burder in constructing these measures. We have more than 1 million cells in Russia at 500 km from the border, and more than 800 thousand cells in Ukraine. Constructing the market potential measure for the cells thus requires computing almost one billion distances and smoothing them. Eberhard-Ruiz and Moradi (2019, p. 263) use $\sigma = 0.042$ as their distance decay for Africa, which lies within our parameter range. They also state that their results are unchanged to doubling or halving the value of their decay parameter.

14 See Appendix A.3 for additional details. We did not compute the network distance for cells. There are many cells (some of them lit) that are not near major roads and for which it is thus hard to compute a meaningful shortest path on the major road network.

$$\ln(\exp_p^{\text{ND}}) = \ln\left(\frac{\min_{j \in \mathcal{B}_P} d_{pj}^{\text{ND}}}{\min_{k \in \mathcal{B}_N} d_{pk}^{\text{ND}}}\right), \quad (5)$$

where $\mathcal{B}_P$ and $\mathcal{B}_N$ are the sets of border crossing points for the positive and negative border segments, respectively.

As fourth and fifth measures, we directly leverage latitude information which capture north–south differences in an obvious way. More precisely, we use the sample mean-centred latitude:

$$\ln(\exp_i^{\text{LAT}}) = \ln(\text{lat}_i / \overline{\text{lat}}). \quad (6)$$

either as a continuous variable or in a discretized form where we divide the border regions into three ‘latitude bands’ – North, Donbass and South.

3.2. First descriptives

Figure 3b shows changes in (log) nighttime lights along the border. We depict changes between 2013 and 2015, that is, 1 year before and 1 year after the start of the conflict. The figure reveals the heterogeneous changes along the border. In particular, lights seem to grow brighter in the south (near the positive border segment in green) and dimmer in the north (near the negative border segment in red). Growth appears especially poor in the Belgorod region, which is only 70 km away from Kharkiv, the closest large city in Ukraine. On the Ukrainian side, the Donbass significantly lost lights (see. e.g., Kochnev, 2019), whereas Crimea significantly gained in lights. The main activity close to Donbass in Russia is concentrated in the Rostov-on-Don area, which does not display a clear positive or negative pattern in NTL changes. One potential explanation is that, although the area is close to the region of armed conflict in the Donbass, it is also close to the southern part that gained better market access to the Crimean peninsula. Another explanation is that the Rostov-on-Don region served as a gateway for firms in the separatist-controlled areas of the Donbass to maintain economic ties and to ship goods to export markets.¹⁵

To provide more formal intuition for how economic outcomes differentially changed along the border in 2014, we run some simple descriptive regressions. We estimate the following two models:

$$\ln(\omega_{it} + 1) = \alpha_i + \delta_t + \sum_{\tau \neq 2014} \gamma_\tau \times D_\tau \times \ln \exp_i + \varepsilon_{it}, \quad (7)$$

¹⁵ Kochnev (2019) provides a detailed account of the banking and trade sanctions that were imposed on the firms in the separatist controlled regions and cites evidence on how these sanctions were evaded.

$$\omega_{pt} = \alpha_p + \delta_t + \sum_{\tau \neq 2014} \gamma_\tau \times D_\tau \times \ln \exp_p + \mathbf{X}_{pt} \beta + \varepsilon_{pt}, \quad (8)$$

where ω_{it} is the NTL value of cell i in year $t = 2006, \dots, 2018$ ¹⁶; ω_{pt} is a dummy variable taking value 1 if plant p exits in year t and 0 otherwise; $\ln \exp_i$ and $\ln \exp_p$ are our pre-conflict measures of exposure; $\mathbf{X}_{pt}$ are plant-level controls; α_i or α_p are cell- or plant fixed effects; and δ_t are year fixed effects. We take 2014 as the start of the conflict and include the years 2006–13 before the annexation and the years 2015–2018 following it. The dummy D_τ equals 1 for year τ and 0 otherwise.

Figure 4 shows estimates of Equation (7) for some of our exposure measures. Panel (a) plots results where we measure a cell's exposure by GDP market potential in Ukraine as in panel (a) of Figure 3. As shown, the coefficients γ_τ were positive before 2013–2014, that is, cells relatively more exposed to Ukrainian market potential grew more pre-conflict. This pattern gets reversed around 2013–2014 in the wake of the violent Euro Maidan protests in Kiev, the annexation of Crimea, and the ensuing armed conflict that erupted in the Donbass: cells relatively more exposed pre-conflict grew relatively less post-conflict. Panel (b) replicates these results using relative GC distance as the cells' exposure measure. As shown, cell-level NTL in Russia was relatively stable before 2013–2014 and the coefficients were positive. In other words, cells relatively farther away from the south (i.e., relatively closer to the north) grew more. This pattern gets again reversed around 2013–2014: nightlights grew more slowly in the north compared to the south.

Taken together, panels (a) and (b) suggest that cells in Russia relatively more exposed to pre-conflict market potential in Ukraine or relatively closer to the negative border segment in the north compared to the positive border segment in the south saw relatively less growth (or more decline) in their lights starting in 2013–2014.¹⁷

Figure 5 shows estimates of Equation (8). Panel (a) plots results where we measure a plant's exposures using the relative GC distance. As shown, the coefficients are negative or zero before 2013–2014, whereas they become significantly positive after 2014.¹⁸ In

16 We add 1 to NTL to deal with cells having zero lights. Intensive-margin estimates, using non-zero cells only, yield similar but somewhat smaller results.

17 The HNTL series combine the DMSP and VIIRS satellite series, using 2013 for purposes of intercalibration. One may thus be worried that the fall in lights is an artifact of a change in series. We do not believe this is a problem because the coefficients we report capture the *differential effect* of exposure to market potential or distance to the border. Any change in the series that is not neutralized by the satellite intercalibration procedure should affect all cells irrespective of their exposure to Ukraine. See Li et al. (2020) for more details on the quality of the series' intercalibration.

18 Panel (a) excludes 2012, which has an abnormally large exit rate. This pattern has been documented before. Iwasaki et al. (2016, p. 169) attribute it to 'to the world economic crisis, whose impact arrived with some time-lag'. While this may be one explanation, we document in Supplementary Appendix F that a mixture of changes in legal enforcement and in the tax environment in 2011–12 are more likely drivers of this observed spike. Note there is some correlation of the increased exit in 2012 with our measure of exposure. As argued in Supplementary Appendix F, this is mostly due to abnormally large exit in the Rostov-on-Don region, located close to the Donbass. While we have no good explanation for why that is the case – since exit was generally higher everywhere in 2012 – we do not think that this poses a serious problem for our subsequent estimates. In Section 5.5, we present evidence from

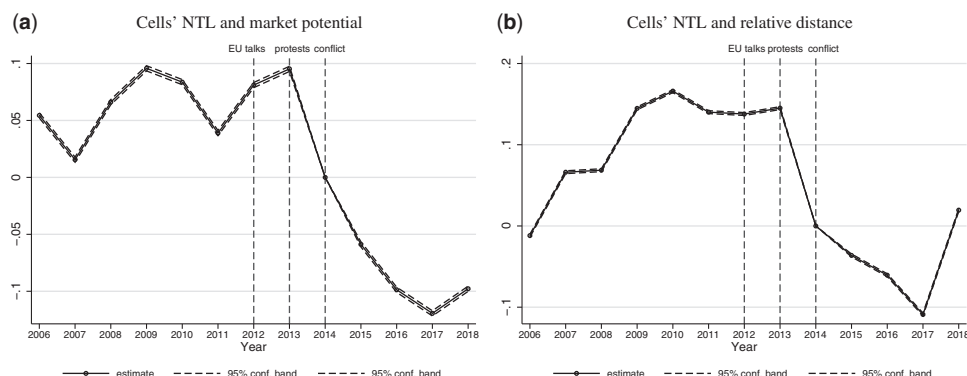


Figure 4. Changes in nighttime lights

Notes: The dependent variable is the log of (one plus) nighttime lights for cell i in year t . Panel (a) depicts the coefficients obtained from Equation (7), where exposure is measured by the log of GDP market potential in Ukraine for cells up to 150 km in Russia. Panel (b) depicts the coefficients obtained from Equation (7), where exposure is measured by the log of GC distance to the southern border relative to the GC distance to the northern border for cells up to 150 km in Russia.

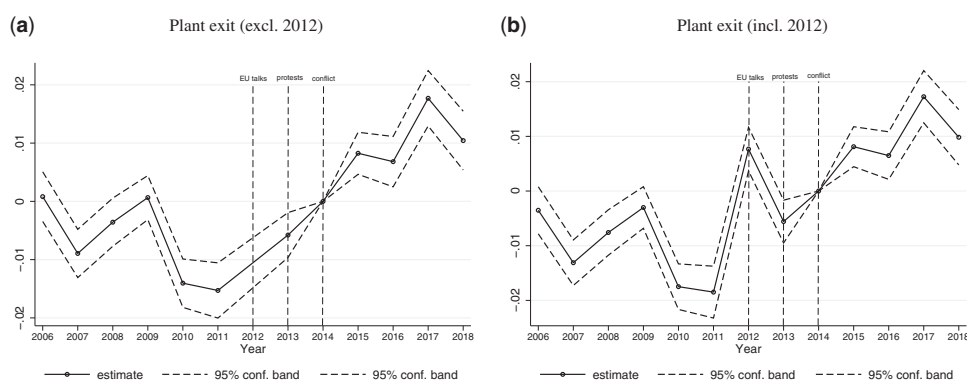


Figure 5. Changes in plant exit

Notes: The dependent variable is a dummy that takes value 1 if plant p exits in year t and 0 otherwise. Panel (a) depicts the coefficients obtained from Equation (8), where exposure is measured by the log of GC distance to the southern border relative to the GC distance to the northern border for plants up to 200 km in Russia. Panel (a) excludes the year 2012 since, as explained in the text and in Supplementary Appendix F, the exit rates show an abnormal pattern in that year. We further control for the minimum distance to the border to isolate the effect of relative position. Panel (b) shows the same figure but includes the abnormal patterns for the year 2012.

other words, while plants closer to the southern border had higher exit rates before 2014, this pattern gets reversed after 2014 with plants relatively closer to the northern border having now higher exit rates.

To summarize, Figures 4 and 5 provide suggestive evidence that economic outcomes in the Russian border regions with Ukraine – as measured by either nighttime lights or

the Central Bank's 'Business Climate Indicator' that suggests exit is not primarily driven by firms' expectations about the worsening relations between Russia and Ukraine.

plant exit – were better for more exposed cells and plants before 2013–2014 but got worse after that date. Put differently, we see a change in trends starting around 2013–2014 in the wake of the Euro Maidan protests in Kiev and the conflict following the annexation of Crimea.

4. EMPIRICAL ANALYSIS

We now turn to our difference-in-differences (DiD) analysis. Ideally, we would observe treated and untreated cells or plants and estimate a standard DiD specification. There is, however, no clear control group in our setting since all cells and plants were treated after 2014 in the wake of western sanctions imposed on Russia. We thus use a different strategy and exploit geographic variation that allows for meaningful comparisons. More precisely, we work within a framework where *observations differ by treatment intensity*, either because they were differently exposed to economic activity in Ukraine before 2014 (absolute measures) or because they were differently affected by changes in border frictions after 2014 (relative measures). We conjecture that observations more exposed to economic activity in Ukraine pre-conflict and observations that experienced a net decrease in market access have worse outcomes – less cell-level NTL growth and more plant-level exit – post-treatment.¹⁹

Figure 6 illustrates the variation we exploit in our data using cells as an example. We depict the ‘negative border’ in the north as the segment N and the ‘positive border’ in the south as the segment P . As shown, our empirical strategy compares less exposed cells i (that received a ‘smaller dose’ of treatment) with more exposed cells i' (that received a ‘larger dose’ of treatment). Exposure depends both on the distance to the border and the relative position along the border. Cells i and i' in Figure 6 are located on the same border-distance isocurve $\bar{d} = \min\{d_P, d_N\}$, but cell i' is more strongly exposed to the negative border N , whereas cell i is more strongly exposed to the positive border P (i.e., $d_P/d_N > d_{P'}/d_{N'}$). We thus anticipate worse outcomes for cell i' compared to outcomes for cell i , conditional on being on the same distance iso-curve. Hence, in our empirical specification, we estimate the effect of the exposure measure conditional on distance from the border. We limit ourselves to a buffer of 300 km from the border, which provides a large enough sample (580,945 cells and 80,287 plants) yet restricts the geographic variation to an area we think allows for meaningful comparisons.

4.1. Cell-level regressions

We first regress our proxies for economic outcomes of cell i in year t on absolute and relative exposure to Ukraine as follows:

19 By construction, our exposure measures are time-invariant so that we cannot disentangle them from cell or plant fixed effects. However, we can estimate their differential effect in the post treatment period.

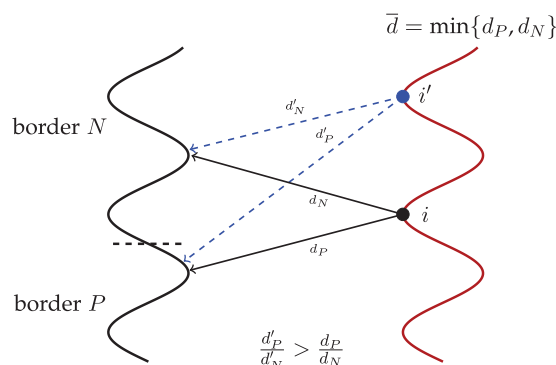


Figure 6. Relative distance along border-distance isocurves

$$\ln(\omega_{it} + 1) = \beta_0 + \gamma_1 \times \text{post}_{2014} \times \text{lnminDist}_i \times 1(\exp_i = \text{relative}) + \gamma_2 \times \text{post}_{2014} \times \ln \exp_i + \alpha_i + \delta_t + \varepsilon_{it}, \quad (9)$$

where ω_{it} is the NTL value of cell i in year t ; post_{2014} is a dummy variable taking value 1 starting in 2014; α_i and δ_t are cell and year fixed effects; and ε_{it} is the error term. The variable $1(\exp_i = \text{relative})$ is an indicator for our relative exposure measures. Thus, in Equation (9) we control for the minimum distance to the border, but only when using a relative exposure measure. The reason is that our absolute exposure measures are highly correlated with distance from the border so that controlling for the minimum distance is not meaningful.²⁰ Our coefficient of interest is γ_2 .

Table 1 summarizes our results. Starting with column (1), we see that cells within 150 km from the border grew less after 2014 than cells within 150–300 km. Hence, lights dimmed on average close to the border in the post-treatment period, consistent with the descriptive evidence from Section 2. Columns (2) and (3) show results for our market potential measures (based on either lights or GDP). As shown, more exposed cells pre-conflict lost after 2014 compared to initially less exposed cells, irrespective of how we measure market potential.

To reveal the different effects of treatment along the border, columns (4)–(7) show results for relative exposure (conditional on the minimum distance from the border). Column (4) splits our distance-band dummy from column (1) into two parts, one for being close to the negative border segment in the north, and one for being close to the positive border segment in the south. As shown, the average effect in column (1) is highly heterogeneous: splitting it into a positive and a negative part reveals that cells close to the border in the north lost luminosity, whereas cells close to the border in the south gained luminosity. Column (5) shows that cells relatively closer to the negative border than to the positive border – as measured by the relative GC distance – grew less post-2014. Columns (6) and (7) show that this continues to hold if we measure

20 The only exception is when we split the distance bands into a positive and a negative part. Although we view this as a relative measure, including the minimum distance is too collinear with our measure.

Table 1. Cell-level changes in NTL before and after 2014

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Absolute exposure			Relative exposure			
	distance band	MP NTL	MP GDP	distance band	GC	LAT	LAT bands
Post-2014 × band	−0.223 ^a (0.002)						
Post-2014 × band(positive)				0.303 ^a (0.005)			
Post-2014 × band(negative)				−0.264 ^a (0.002)			
Post-2014 × Lat(Donbas)							−0.271 ^a (0.003)
Post-2014 × Lat(North)							−0.246 ^a (0.003)
Post-2014 × exposure		−0.166 ^a (0.001)	−0.159 ^a (0.001)		−0.171 ^a (0.001)	−0.026 ^a (0.000)	
lnminDist included					✓	✓	✓
Cell FE	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓
Observations (cells×years)	8,133,230	8,133,230	8,133,230	8,133,230	8,133,230	8,133,230	8,133,230
R ²	0.695	0.696	0.696	0.696	0.698	0.696	0.697

Notes: The dependent variable is $\ln(\omega_{it} + 1)$. All regressions include cell and year fixed effects. Standard errors are clustered at the cell level. band is a dummy with value one if the cell is less than 150 km from the border, whereas band(positive) is a dummy with value one if the cell is less than 150 km positive border, and is closer to the positive border than to the negative border. band(negative) is constructed in the same way, but for the negative border. We include all cells up to 300 km from the border. exp is our exposure measure as indicated in the column header (GC = great circle distance (4); LAT = centred latitude, (6); MP Lights and MP GDP = market potential based on either raw NTL (3) or using cell-level GDP (A.2)). Estimates in columns (5)–(7) include the minimum distance from the border. ^a $p < 0.01$, ^b $p < 0.05$, ^c $p < 0.1$.

relative exposure by the cell’s latitude, either continuously or by using latitude bands. We further see that both the area close to the Donbass and the north saw less growth in lights post-2014 than the south, with a stronger negative effect in the Donbass latitude band.

Overall, our DiD estimates for absolute exposure show that cells more exposed to Ukraine pre-conflict saw a dimming in their lights compared to less exposed cells. Our estimates for relative exposure further show that this effect varies with treatment intensity along the border as cells located closer to the negative border segment relative to the positive border segment experienced a dimming in their lights compared to cells located closer to the positive border segment relative to the negative border segment. In other words, the different changes along the border induce different changes in economic activity, as measured by nighttime lights.

How large are the post-2014 effects of changes in nighttime lights on GDP? Figure 7a illustrates the magnitude of the economic effects of the different exposure measures on nighttime lights using the predicted changes in the dependent variable at the 1st, 5th and 10th exposure deciles, respectively. On average (across exposure measures), cell-level NTL increased by about 33% in the post-2014 period at the 1st decile, whereas it decreased by about 10% at the 10th decile: hence, lights increased by about 43% less in the post-treatment period for the most exposed cells compared to the least exposed cells. To translate our estimates into GDP equivalent figures, we require estimates of the elasticity of GDP with respect to nighttime lights. In Appendix C, we follow Henderson *et al.* (2012) and Hodler and Raschky (2014) and estimate the elasticity of regional per capita GDP to regional average nighttime lights to be around 0.2 in Russia; the long difference estimates between 1996–1997 and 2017–2018 are 0.17. Thus, a 43% difference in NTL changes for the most exposed cells corresponds approximately to a 6.2% to 7.1% difference in GDP changes, depending on the elasticity.²¹ Table 2 provides a summary of the economic effects of our estimates based on different values of the elasticity of GDP to nighttime lights from the literature. As shown, we find effects somewhere in between 6% and 12%, depending on the assumed values of the elasticity.

4.2. Plant exit regressions

We next run regressions using plant exit as the dependent variable. More precisely, we estimate the following model:

$$\omega_{pt} = \beta_0 + \gamma_1 \times \text{post}_{2014} \times \ln \text{minDist}_p \times 1(\exp_p = \text{relative}) \\ + \gamma_2 (\text{post}_{2014} \times \ln \exp_p) + \mathbf{X}_{pt} \beta + \alpha_p + \delta_t + \varepsilon_{pt}, \quad (10)$$

where ω_{pt} takes value 1 if plant p exits in year t , and value 0 otherwise; $\ln \exp_p$ is one of our exposure measures from Section 3; $\mathbf{X}_{pt}$ are (time-varying) plant-level controls; and α_p and δ_t are plant and year fixed effects. The variable $1(\exp_p = \text{relative})$ denotes as before an indicator for relative exposure measures. We estimate Equation (10) using a linear probability model, but probit regressions yield very similar results.

Table 3 shows our results. Columns (1)–(3) report results for our absolute exposure measures. As can be seen, we find no effects for our market potential measures, and small negative effects for the 150 km distance band. Figure 3a shows that the regions with the largest pre-conflict exposure to Ukraine are both north and south of the Donbass. In other words, there is no clear north–south pattern in exposure as measured by market potential, which may explain the absence of effects in columns (2) and (3). Columns (4)–(8), however, reveal that relative exposure is positively associated with a

21 Henderson *et al.* (2012) report 0.33 and 0.31 for their elasticities, and Hodler and Raschky (2014) find 0.23 and 0.39, respectively.

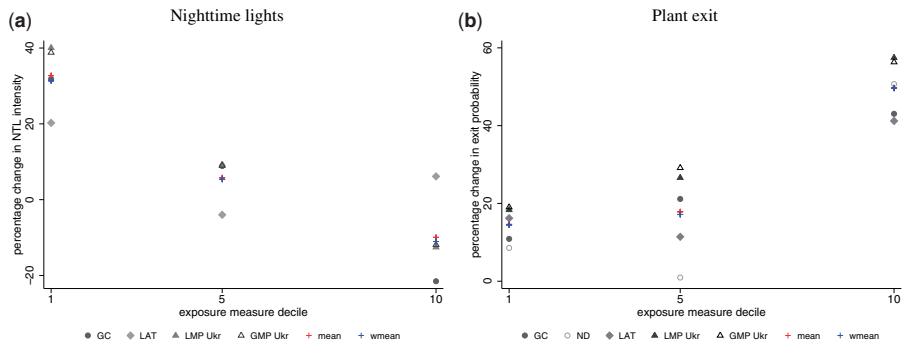


Figure 7. Predicted post-treatment changes at the 1st, 5th and 10th exposure deciles

Notes: Scatter plots of the average change in (a) cell-level NTL and (b) the exit probability in each decile post-2014 compared to pre-2014. Each decile reports the distribution of the increase for our continuous exposure measures. ‘Exposure measure decile’ 5, for example, depicts the distribution of the average change in NTL or in the exit probability for cells or plants with exposure between the 40th and 50th percentiles for each of the continuous exposure measures in Table 1 and Table 3. We also report the simple mean across measures, as well as a weighted mean where the weights are the inverse of the estimated standard deviations.

Table 2. Magnitude of GDP changes, based on different elasticities

	Average difference $\hat{y}$ 1st–10th deciles	Our estimated elasticities		Hodler and Raschky (2014)		Henderson <i>et al.</i> (2012)	
		ϵ_{LT}	ϵ_{ST}	ϵ_{LT}	ϵ_{ST}	ϵ_{LT}	ϵ_{ST}
		0.17	0.2	0.23	0.39	0.33	0.31
Implied changes in GDP based on the elasticity							
Raw NTL (umean) (%)	42.70	6.20	7.12	7.98	11.98	10.59	10.10
Raw NTL (wmean) (%)	42.39	6.16	7.07	7.93	11.89	10.52	10.03
GDP using lights (umean) (%)	34.44	5.85	6.89	7.92	13.43	11.37	10.68
GDP using lights (wmean) (%)	33.28	5.66	6.66	7.65	12.98	10.98	10.32
Simple average (%)		5.97	6.93	7.87	12.57	10.87	10.28

Notes: The unweighted mean (umean) is computed across our exposure measures per decile. The weighted mean (wmean) is computed across exposure measures per decile and weighted using the inverse of the predicted values’ standard deviations per decile. As explained in Appendix A.1, the magnitude of the effects for nighttime lights need to be adjusted by $1/(1 + \epsilon)$ to make them comparable to the estimates using lights-apportioned GDP instead of raw lights.

higher probability of plant exit after 2014. Put differently, plants more exposed to the negative border are more likely to exit post-2014 than less exposed plants. This finding holds for all our relative exposure measures and is thus robust. The only a priori unexpected result is the insignificant interaction term between post-2014 and the Donbass latitude band in column (8). This result suggests that plants in the Donbass latitude band were not more likely to exit post-2014, which may reflect the ambiguous border changes in the conflict region of the Donbass and the fact that this area is between the

Table 3. Plant-level changes in exit probability, before and after 2014

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Absolute exposure			Relative exposure				
	distance band	MP NTL	MP GDP	distance band	GC	ND	LAT	LAT bands
Post-2014 x band	−0.008 ^a (0.002)							
Post-2014 x band(positive)				−0.030 ^a (0.004)				
Post-2014 x band(negative)				−0.005 ^a (0.002)				
Post-2014 x Lat(Donbas)								0.003 (0.003)
Post-2014 x Lat(North)								0.026 ^a (0.002)
Post-2014 x exposure		0.001 (0.001)	−0.000 (0.001)		0.012 ^a (0.001)	0.005 ^a (0.001)	0.003 ^a (0.000)	
Plant controls	✓	✓	✓	✓	✓	✓	✓	✓
lnminDist included					✓	✓	✓	✓
Plant FE	✓	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓	✓
Observations (plants×years)	528,147	528,147	528,147	528,147	528,147	528,147	528,147	528,147
R ²	0.222	0.222	0.222	0.222	0.222	0.222	0.222	0.222

Notes: The dependent variable is a dummy with value 1 if plant p exits in year t , and 0 otherwise, band is a dummy variable with value 1 if the plant is less than 150 km from the border, whereas band(positive) is a dummy with value 1 if the plant is less than 150 km from the positive border, and it is closer to the positive border than to the negative border. band (negative) is constructed in the same way, but for the negative border. We include all plants up to 300 km from the border. exp is our exposure measure as indicated in the column header (GC = great circle distance (4); ND = network distance (5); LAT = centered latitude, (6); MP Lights and MP GDP = market potential based on either raw NTL (3) or using cell-level GDP (A.2)). Estimates in columns (5)–(8) include the minimum distance from the border. ^a $p < 0.01$, ^b $p < 0.05$, ^c $p < 0.1$.

north and the south and thus may be on net little affected by the relative border changes (see [Figure 1b](#)).²²

How large is the post-2014 effect on the exit probability of plants? We first compute the raw exit rate over the 2006–13 period, which is 7.02%.²³ As is known (e.g., [Iwasaki et al. 2016](#)), yearly exit rates are much lower in Russia than in more developed countries and this clearly shows in our data. [Figure 7b](#) graphically illustrates the magnitude of the economic effect of the different exposure measures on plant exit using our predicted exit probabilities at different exposure deciles. On average, the exit probability increased by about 15% at the 1st decile, whereas it increased by about 50% at the 10th decile. Thus, the most exposed plants have a 35% larger increase in their exit probability post-2014. With respect to the 7.02% baseline rate, this corresponds to a 2.5 percentage points increase at the top decile, but only to a 1 percentage point increase in the bottom decile. As for nighttime lights, given the baseline exit probabilities these are sizeable numbers which show that the 2014 conflict had effects of quite different magnitude depending on the plants' initial exposure to Ukraine and exposure to border changes.

Contrary to the case of cells, which are stable over time, the composition of our sample in terms of plants changes over time with entry and exit. One may thus wonder how much of the above effects are driven by unobserved plant characteristics that change with sample composition and how much directly by changes in the borders after 2014. To get a rough idea, we predict the exit probabilities by only including the plants that were active in 2013 in our sample and by not including the plant fixed effects in the prediction. [Supplementary Appendix Figure D4](#) shows the results. The gap between the 1st and the 10th deciles now shrinks to 0.6 percentage points. This suggests that about 70% of the effect we find is driven by compositional changes (in unobserved plants-level characteristics) and about 30% by changes in exposure. In words, plants after 2014 had unobserved characteristics that made them more likely to exit compared to similar plants before 2014, irrespective of exposure, which may be a result of sanctions or other changes in the environment after 2014.

4.3. Robustness checks

We run a large number of robustness checks, which we briefly summarize here. Most tables and additional details are relegated to [Supplementary Appendix D.2](#).

²² While the Ukrainian government remained in control of the border in the north and made it less permeable – and completely lost control of the border in the south – the border in the contested region was controlled by no one perfectly. [Kochnev \(2019\)](#) cites evidence suggesting that firms in the separatist controlled areas of the Donbass, following the trade restrictions with the rest of Ukraine and the disruption of access to banking, partly shipped goods to outside markets using intermediate firms located in the area bordering the Donbass in Russia. Furthermore, there was a substantial influx of refugees into that regions. In a nutshell, whether this affected firms in the region positively or negatively is unclear, but our estimates do not reveal more exit post 2014.

²³ [Supplementary Appendix Table D.3](#) reports the predicted exit rates pre- and post-2014 by exposure deciles (1st, 5th and 10th deciles, respectively) and shows that they vary systematically with pre-conflict exposure.

First, instead of using the relative distance to the borders – conditional on overall minimum distance to the border – we estimate specification (10) using separate minimum GC and NDs to the positive and the negative border segments, respectively. [Table 4](#) shows that the two measures have separately the expected effect and are individually significant in most of our estimations.

Second, we re-estimate [Table 1](#) with cell-level GDP apportioned using lights as the dependent variable instead of raw lights. [Supplementary Appendix Table D.4](#) shows the results. They are qualitatively similar. As shown in [Table 2](#), the implied magnitudes of the effects are also in the same range as those using raw lights.

Third, we exclude all cells that were unlit all the time before 2014 and became lit afterwards. The results in [Supplementary Appendix Tables D.5 and D.6](#) show that our results are robust to the exclusion of such cells. We also ran regressions where we exclude all cells that had lights of 0 or 1 in 2013, which gives similar results to our baseline estimates with marginal changes in the coefficients' magnitudes.

Fourth, we estimate [Equation \(10\)](#) with industry-year fixed effects instead of plant fixed effects. In doing so, we control for: (1) whether the plant belongs to a multi-unit firm; (2) the average population of the municipality the plant is located in; (3) a dummy taking value 1 if the plant is located in a big city and 0 otherwise; and (4) the plant's average exposure to market potential in Russia computed as in [Equation \(3\)](#) in Section 3. [Supplementary Appendix Table D.7](#) shows the results. We find that after 2014, plants initially more exposed to market potential in Ukraine had a slightly higher exit probability than less exposed plants.

Fifth, we run regressions where we include only the 90% of plants that 'truly exit', that is, we exclude plants that exit because of accession and mergers. [Supplementary Appendix Table D.2](#) shows that our coefficients of interest are unaffected by these changes.

Last, we verified that none of our results change if we use heteroscedasticity robust standard errors instead of clustered ones.

5. THE ROLE OF MARKET ACCESS

We now provide additional evidence which suggests that differential changes in market access drive differential changes in economic outcomes. We first explain why access to the Crimean market seems important for the southern Russian regions. We then provide additional analyses to rule out other factors than market access that might confound our estimates. We relegate detailed results to [Supplementary Appendix E](#) to save space. We then leverage, in a final Section 6, new data we assembled to show that the closing of border crossings explains some of the local variation in economic outcomes we see in the northern border regions (Bryansk, Kursk, Belgorod and Voronezh).

5.1. The importance of the Crimean market

Crimea's average population, including Sevastopol, for 2011–2013 was 2,296,290, and its GDP was about 6.5 billion USD in nominal prices. By comparison, the close

Table 4. Changes in NTL and plant exit before and after 2014, separate distance measures

	Cell-level NTL and GC			Plant exit and GC			Plant exit and ND		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Post-2014 × ln dist pos	−0.183 ^a (0.002)		−0.029 ^a (0.002)	0.016 ^a (0.001)		0.017 ^a (0.002)	0.006 ^a (0.001)		0.004 ^a (0.001)
Post-2014 × ln dist neg		0.730 ^a (0.006)	0.674 ^a (0.008)		−0.030 ^a (0.003)	0.003 (0.005)		−0.020 ^a (0.002)	−0.012 ^a (0.003)
lnminDist included	✓	✓	✓	✓	✓	✓	✓	✓	✓
Cell or plant fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	8,133,230	8,133,230	8,133,230	528,147	528,147	528,147	528,147	528,147	528,147
R ²	0.697	0.699	0.699	0.222	0.222	0.222	0.222	0.222	0.222

Notes: The dependent variables are: $\ln(\omega_{it} + 1)$ in columns (1)–(3); and a dummy that equals 1 if a plant exits in year t and 0 otherwise in columns (4)–(9). The sample includes all cells or plants within 300 km from the border. Standard errors are clustered at the cell or plant level. The positive and negative distances to the border are measured by either the GC distance in columns (4)–(6) or network (ND) distance in columns (7)–(9). Columns (4)–(9) include plant age as a time-varying control. ^a $p < 0.01$, ^b $p < 0.05$, ^c $p < 0.1$.

southern Russian region to Crimea on the mainland – Krasnodar Krai – had an average pre-conflict population of 5,310,593 and average GDP about 48.5 billion USD. Thus, Crimea had about 40% of the population of Krasnodar and about 31% of its GDP per capita. This represents a substantial potential increase in market size for the Krasnodar region after the annexation.

To gauge the importance of the Crimean market for the south, we can leverage trade statistics. [Figure 8\(a\) and \(b\)](#) shows that Crimea's foreign trade changed significantly between 2010 and 2017.

Both imports and exports with the rest-of-the-world literally disappeared after the annexation, meaning that Crimea – including Sevastopol – now trades almost exclusively with the Russian Federation (which is classified as 'domestic trade').²⁴ Overall, [Figure 8\(a\) and \(b\)](#) suggests that the Crimean market has become important for Russia as both an export destination and an import source. Given the well-established role of gravity in shaping trade patterns, the Crimean market should be more important for nearby Russian regions such as Krasnodar Krai, especially after the completion of the bridge over the strait of Kerch in 2018 which provides a land route to Crimea.

Is it the market potential of Crimea that directly matters, or the possibility to indirectly access other markets via Crimea? We think there is no reason that the latter should be the case. We have seen that foreign trade of Crimea almost vanished. Furthermore, the port of Novorossiysk on the Russian mainland in Krasnodar Krai has direct access to the Black Sea and is one of the largest merchandise ports in Russia (see [Figure 1a](#)). Although much attention has been given to the ports of Crimea in the discussion of their strategic role for Russia, that strategic role is military: the Black Sea fleet historically has its main base in Sevastopol, and the naval bases were much more important to Russia than the commercial ports in that region.

5.2. Sanctions after 2014

A first potential confounder may be the imposition of mutual sanctions between an alliance of Western countries and Russia in the aftermath of the Crimean annexation. It is a priori unclear whether these sanctions – which targeted specific industries and groups of countries on the Western and on the Russian sides – may have differential north–south effects that could confound our DiD estimates. Since the sanctions targeted specific product groups they might have had a disproportionate impact on regions with a large share of firms in the sanctioned industries, and those may be located in the north.²⁵

24 See also [Komissarova et al. \(2018\)](#) for a discussion and figures on Crimean trade before and after 2014. Korovkin and Makarin (2023) provide additional details on the changes in trade relationships between Russia and Ukraine and, within Ukraine, between regions with different ethnic composition.

25 For a full list of sanctioned/embargoed goods, as well as the countries supporting them, see [Bélin and Hanousek \(2021\)](#) and [Crozet and Hinz \(2020\)](#). Note that there were mutual sanctions: Western countries sanctioned the export of specific goods to Russia, and Russia retaliated by restricting the imports of other, mainly agricultural, goods from Western countries.

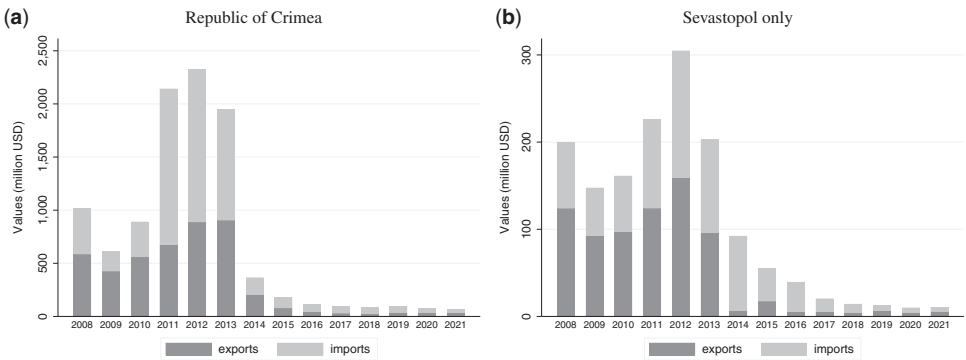


Figure 8. Total exports and imports for the Republic of Crimea and the city of Sevastopol

Notes: Total value of exports from and imports to Crimea. Panel (a) depicts trade figures for the Republic of Crimea. Panel (b) depicts trade figures for the city of Sevastopol. The trade data sources are UkrStat (ukrstat.gov.ua) before 2014, and CrimeaStat (crimea.gks.ru) from 2014 onwards.

To address this concern, we construct an exposure measure to sanctions at the municipal district level (see [Appendix A.3](#) for details). We then re-estimate our baseline models for cells and plants in [Tables 1](#) and [3](#) including our constructed measure of exposure to sanctions interacted with the post-2014 dummy to see how cells and plants in municipalities more exposed to trade sanctions in the pre-treatment period fared differently after the conflict and the imposition of the sanctions. In all regressions, the baseline exposure effects are of similar magnitude and significance, that is, accounting for the sanctions does not change our original estimates (see [Supplementary Appendix Tables E.11–E.12](#)). The effect of the exposure to sanctions post-2014 is positive for cells conditioning on the absolute exposure measures, whereas it is negative conditioning on the relative exposure measures and controlling for the log of the minimum great-circle distance. For plant exit, the inclusion of the sanctions control shows no significant effect on plant exit with industry–year fixed effects (not reported), and weakly negative effects in regressions with plant fixed effects (see [Supplementary Table E.12](#)). Our coefficient of interest, that is, $\text{post-2014} \times \text{exposure}$, remains largely unchanged. Overall, we find weak regional effects of sanctions which do not affect our estimates regarding the importance of market access to Ukraine for the Russian border regions.

5.3. The 2014 winter Olympic games in Sochi

The 2014 Winter Olympic Games in Sochi are a second potential confounder. Sochi is a city on the Black Sea in the Krasnodar region, about 300 km from the southern Russian–Ukrainian sea border (see [Figure 1a](#)). It hosted the 2014 Winter Olympic Games. One concern is that the geographic orientation – favouring the south – and the timing – just before the annexation – overlap with the two key dimensions we exploit to identify the causal effect of market access. Hence the Olympics could explain the better

performance of the southern regions, especially since they had an explicit ‘regional development’ dimension (see, e.g., Müller, 2014).

We have investigated this problem and show below that our results do not seem driven by the Olympics. First, Figures 4 and 5 show that the differential economic performance prior to 2014 favoured the north yet got reversed after 2014. Thus, if there is a ‘Sochi effect’ prior to 2014, removing it would imply that the change in trend we see after 2014 is even more pronounced. Second, for a ‘Sochi effect’ to bias our estimates in the wrong direction, we would need that the growth post-2014 due to the Winter Olympics be much stronger than the growth pre-2014, which runs counter the empirical (and anecdotal) evidence discussed in Müller (2014) and Firgo (2021).²⁶ The post-2014 Olympic effect – if there was one at all – seems to have been weak and short lived.

To check more formally that it is not public spending on the Winter Olympics that drives our results (or support to firms in view of the Olympics), we run various regressions where we include controls for the distance to Sochi. We compute the shortest (GC) distance to Sochi from all cells and plants in our sample. In our preferred specification, we estimate a model where we control for cell or plant distances from Sochi within 0–50 km, 50–100 km and 100–150 km distance bands (see Supplementary Appendix Tables E.13–E.14). All our main results survive this new specification although – as expected – the magnitude of some coefficients decreases.²⁷

The results in Supplementary Appendix Table E.13 show that after 2014, cells close to Sochi grew faster than cells located further away, but the main results for our exposure measures remain quantitatively and qualitatively unchanged: even conditional on distance from Sochi, cells more exposed to Ukraine in the south grew more than less exposed cells. We repeat this exercise with plant exit in Supplementary Appendix Table E.14. We find more plant exit up to 100 km from Sochi than their more distant counterparts after 2014. Again, the inclusion of additional Sochi controls does not alter the results for our exposure measures, and all coefficients are qualitatively the same: plants more exposed to Ukraine pre-2014 tended to exit more post-2014, conditional on their distance from Sochi. We thus believe it is fair to say that Sochi does not explain our main results.

As expected, our market potential exposure results for plant exit remain insignificant, as in the baseline regressions. However, the (unreported) results for using industry-year

26 Firgo (2021) documents that Olympic Winter Games generally have no specific positive long-run effect on host regions, but in the eight years between attribution and hosting there is a positive growth effect for regions in terms of GDP per capita. For the 2014 winter Olympics, this effect appears to have been sizeable. As explained, this would only reinforce the change in trend we see.

27 We also experimented with a continuous distance measure for all cells from Sochi, but in that case we obtain a variable that is too collinear with our exposure measure and that is very likely to soak up the variation due to exposure to Ukraine. We also think it makes little sense to assume that there is a ‘Sochi effect’ that stretches along the whole border with Ukraine. We are thus more confident in our results using the distance bins from Sochi up to 150 km.

fixed effects yield significantly positive coefficients, meaning that plants more exposed to Ukraine pre-2014 tended to exit more after the onset of the conflict.

5.4. Public investments and subsidies

A third potential set of confounders may be subsidies to private or state-run businesses, which target more the south than the north. There might be politically motivated investments to support private firms, but we could not find evidence for this. We know about a free economic zone that has been established in the territories of the Republic of Crimea and the city of Sevastopol by the Federal Law of 29 November 2014.²⁸ This free economic zone seems, however, both relatively small and is not on the Russian mainland and thus does not show up in the data used for our estimations.

Another politically motivated support is related to import substitution policy, investments into which amount to more than 3 trillion rubles from 2015 to 2021, with direct government funding totalling more than 500 billion rubles according to the Federation Council Committee on Economic Policy. Import substitution programmes aim to reduce dependencies on imports especially in the agricultural, machinery and IT sectors. The former might have had a disproportionate impact on the southern regions – Krasnodar and Rostov – which traditionally have substantial agricultural production. The trade embargoes on food products, introduced in 2014 as retaliation to Western sanctions, may have boosted the growth of the agro-industrial complex. Yet, [Shagaida and Uzun \(2016\)](#) rather attribute the growth in the agricultural sector post-2014 to the devaluation of the ruble, which created favourable conditions for more domestic production. We have seen in Section 5.2 that there was more growth in lights after 2014 in municipalities more exposed to sanctions, yet weak effects on manufacturing plant exit. This may be due to the positive effect of import sanctions on the agro-industrial production. However, the link between agriculture and lights is usually weak, and our baseline estimates do not change much when controlling for sanctions.

Turning to public investments, substantial investments were made in Crimea, especially to connect its energy and transportation grids to those in Russia. First, there is the ‘Power bridge’ to Crimea, which are cables, overhead power lines and substations built to connect the Crimean energy system to the energy system of the Russian Federation. Its construction started in June 2014, and was completed in 2016, for a total cost of about 800 million dollars. Second, the bridge to Crimea over the strait of Kerch was built, which provides road and rail access to Crimea. Its construction started in 2016 and was completed in 2018 for a total cost of about 4 billion dollars.²⁹ These two large public investment projects to link Crimea to the Russian power and transportation grid

28 Available online at https://www.economy.gov.ru/material/file/d4cae5f3397494aafc525c3720e8eb0b/report_2020.pdf (last accessed on 7 February 2023).

29 Since the opening of the bridge for exploitation in 2018, it has been crossed by more than 21 million automobiles, more than 3 million trucks, and 233 thousand buses. The recorded number of crossings

may partly explain the growth in lights and the higher survival of plants in the southern Russian regions if they benefitted from the extra economic activity generated by these constructions. We do not view this as problematic for our estimates since the activity generated by these projects is directly linked to the new access to the Crimean market. In other words, we would not have seen this beneficial effect in the south had the annexation not occurred and access to serve the Crimean market become more important. In that sense, these are economic benefits generated by market access considerations. However, it is clear that these cannot be considered as being economic benefits generated by free market forces. Hence, if they drive most of the results we estimate we need to be cautious when extrapolating our results to other settings and countries.

5.5. Expectations about future conflict

A last confounder, similar to one in [Redding and Sturm \(2008\)](#), is people's expectations in 2014 about a future deterioration of the relationship with Ukraine. Although this is a possibility, it is hard to see why this would affect more expectations in the north than in the south. It is also hard to measure, but we can proxy it using public opinion polls.

The state-owned polling institute Russian Public Opinion Research Centre (WCIOM) and the non-governmental polling organization Levada Centre (which has since been classified as a foreign agent) both reveal a similar picture of public opinion among Russians about their expectations concerning the future development of the conflict between Russia and Ukraine. On 5 November 2014, WCIOM published results stating that 'twice as many respondents say that the situation in Ukraine remains unchanged today as in August of this year (50% and 23%, respectively). At the same time, the proportion of respondents who believe that the situation is worsening significantly decreased (from 72% to 38%)'. Levada conducted a survey on 22–25 August 2014, which shows that 'Russians however believed that it was unlikely that the tensions between Russia and Ukraine would escalate into a major war. Close to 6 out of 10 respondents (59%) did not believe, in 2014, that there is a war between the two countries'. To summarize, it does not seem that there was a general feeling that escalation of the situation was imminent and unavoidable. Unfortunately, we do not have more detailed opinion surveys on the border regions' populations, so that these results should be taken with a grain of salt. People close to the border may perceive things differently than in Moscow, St. Petersburg, or the far-away regions in the east. However, Korovkin and Makarin (2023, p. 23) find that 'attitudes of Russian citizens towards Ukraine did not change differentially across Russian regions with different shares of ethnic Ukrainians'. Since border regions close to Ukraine have a larger share of ethnic Ukrainians, it is thus unlikely that people along the border were more pessimistic than anywhere else.

per day in both directions in 2020 was 37,374 vehicles. This reinforces our idea that Crimea has become an important market and destination for firms and people in the southern border regions.

A final piece of suggestive evidence comes from a measure of firms' expectations about the development of the economy using the Business Climate Indicator (BCI). This indicator is computed on a monthly basis by the Central Bank of Russia based on a large representative survey of about 14 thousand non-financial firms. The questions and answers provide a good view of firms' perceived operating conditions, taking into account a large range of internal and external factors. The BCI reflects factual and expected changes in production and demand, where expectations are the firms' assessments on the dynamics over the next 3 months. A 1 month lead of this indicator shows a fairly high correlation with GDP and provides thus a good forecast of expected future changes in the economy. [Supplementary Appendix Figure E5](#) shows monthly changes in the BCI from 2004 to 2022. We clearly see three sharp drops: in the midst of financial crisis of 2008; at the onset of the coronavirus disease 2019 (COVID-19) pandemic in 2020; and at the beginning of the 'special military operation' in Ukraine in 2022. Observe that expectations about the business climate were rather stable around 2012 and start to decrease in late 2013 following the Euro Maidan. However, we do not see a sharp drop anywhere near the level of the financial crisis, COVID-19 or the 2022 invasion.

5.6. Disruptions in cross-border electricity trade

We finally investigated whether disruptions in cross-border electricity trade after 2014 could drive our results. Lights obviously require electricity, so that disruptions could explain changes in luminosity. These disruptions could also affect more strongly firms that are energy intensive. Unfortunately, the data on power grids are sparse and hard to use. We ran some regressions creating a variable 'distance to main power line' but did not find any effects.³⁰ We also created a variable at the two-letter industry–region level to proxy the industries' energy intensity. We included post-2014 $\times$ log of average electrical energy consumption over 3 years before the conflict at the plant's industry–region level into our baseline regressions for plant exit. The baseline results continue to hold and there is no effect of this additional control on plant exit, that is, plants in industry–regions that are more energy-intensive before the conflict did not exit more in the post-treatment period.

6. MORE EVIDENCE ON MARKET ACCESS: THE LOCAL EFFECTS OF CLOSED BORDER CROSSINGS

Having ruled out possible confounders, we now present additional evidence for the role of market access. To this end, we leverage a novel dataset we constructed on local border

30 We used the estimated world energy grid (see [Arderne et al., 2020](#)) to compute the shortest distance from the cells (plants) to the nearest cross-border energy point. This new measure inherits the locations of the main roads and settlements near the border and accurately captures medium-voltage networks in about 70% of cases. Therefore, it is hard to attribute the effect using that variable solely to proximity to energy networks.

crossings and exploit changes in them over time. Our analysis is restricted to the northern border regions as the border tightening was more stringent and enforced there (recall that the border was no longer under the control of the Ukrainian government in the regions bordering the Donbass and in Crimea). More precisely, we focus on the Belgorod, Kursk, Bryansk and Voronezh regions at a latitude above the Donbass (see [Figure 1](#)).

6.1. Context and importance of cross-border interactions

In 1995, Russia and Ukraine signed an inter-governmental agreement about co-operation of border regions to facilitate cross border commuting. According to that agreement, residents of border regions could cross the border at *local border crossings* in the region they were residents of.³¹ They could stay on the territory of the neighbouring state only within the region into which they entered through the local border crossing, be it for private or for work reasons. The commercial import or export of merchandise at these local border points was prohibited, that is, trade did not occur through these points but had to flow through *international border crossings* only.

In the wake of the conflict, starting March 2015, the Ukrainian government shut down all local border crossings and only kept the international border crossings open. Consequently, residents in some areas had to travel much longer distances to cross the border for work or shopping, and plants and shops in some areas saw substantial changes in their potential access to workers and customers. We exploit this variation to investigate whether: (1) nighttime lights intensity in cells that experienced an increase to the nearest border crossing grew less and (2) plants in zones that experienced an increase in distance to the nearest open border crossings saw their exit probability increase compared to other plants. [Figure 9](#) illustrates the changes in distance to the closest open border crossing in the wake of the conflict. As shown, the need to travel farther to the nearest open international border crossings could substantially increase the road distance in some locations.

We now provide evidence for the intensity of cross-border interactions between Russia and Ukraine.³² We first document the importance of the Ukrainian workforce in

31 We provide a detailed description of the local and international border crossings and our data sources in [Supplementary Appendix H](#).

32 Direct evidence is rare so that most of it is anecdotal. [Kolosov et al. \(2016, p. 395\)](#) is one study of the intensity of cross-border interactions between Russia and other post-Soviet countries between 2010 and 2014. They document that the border with Ukraine is one of the most permeable and widely used before the conflict starts: 'On the border of Ukraine a high intensity of citizen movement was replaced in 2014 by a sharp drop along with a concomitant increase in barrier function. Under the influence of the crisis in bilateral relations, the number of border crossings via road and rail checkpoints decreased almost twofold. As well, the largest reduction in flow occurred in Belgorod and Kursk oblasts, including via the busiest road checkpoints of Troebortnoe-Bachevsk (by 60%) and Hoptovka-Nekhoteevka (by 30%). Meanwhile, at many checkpoints in Rostov oblast, the number of border crossings in 2014 increased, primarily due to refugees'. [Zayats et al. \(2017\)](#) provide an informal discussion and case study for the potential importance of crossborder labour flows. They explain how the conflict has led to a significant decrease in the trans-border intensity of worker movements.

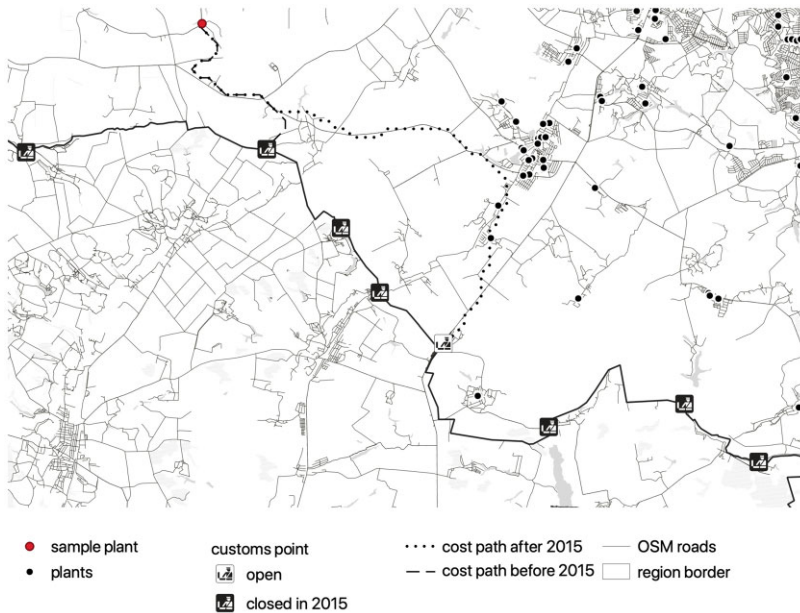


Figure 9. Closure of local border crossings and changes in distance travelled

Notes: Illustration of changes in least-cost paths before and after the closure of local border crossings in 2015. Authors' computations using the OSM road network and 500×500 m grid cells.

Russia and show that it dropped substantially after 2014. Figure 10a shows statistics on the number of Ukrainians with a valid work permit in Russia.³³ The number declines very strongly after 2014. Of course, since these are national statistics – no detailed regional statistics are available – they include Ukrainians working in Moscow or in far-away places like the Siberian oil fields. However, gravity for trade, migration and commuting flows suggest that more Ukrainians should work closer to Ukraine and that the decrease should be stronger in the border regions.

A second way to trace changes in cross-border labour flows is trans-border payments. We approximate the importance of Ukrainian workers in Russian firms by using the total amount of remittances sent from residents and non-residents to Ukraine. According to Russian legislation, foreign workers – whether they are residents or not – cannot receive their salary in cash and must open a bank account labelled in rubles. Workers who temporarily work in Russia or workers who cross the border but live in Ukraine must remit their earnings via the banking system. Hence, changes in remittances capture quite accurately changes in the importance of the Ukrainian workforce in Russia.

Figure 10b depicts the cross-border money transfers made by foreigners (with or without resident status in Russia) in favour of residents and non-residents in Ukraine,

33 According to the Federal Migration Service, there were 1.6 million Ukrainians who came legally or illegally to Russia to earn money. These were mostly residents of the eastern regions of Ukraine.

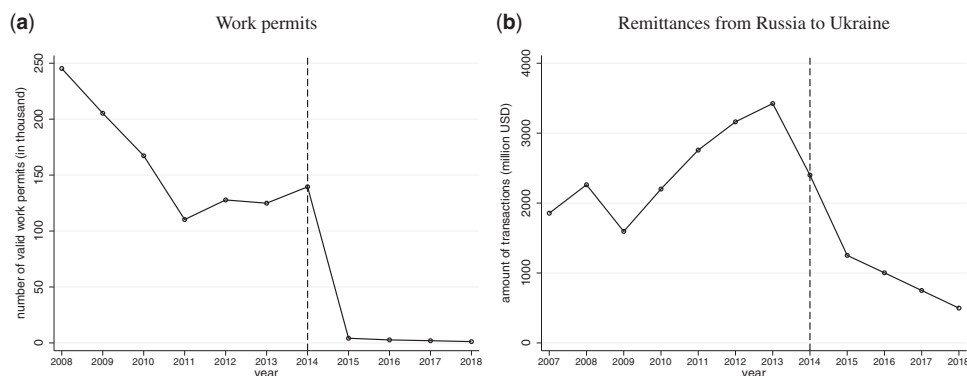


Figure 10. Number of Ukrainians with valid work permits and amount of remittances

Notes: Panel (a) depicts number of Ukrainians with a valid work permit in Russia. Panel (b) depicts the total amount of money remitted to Ukraine.

with and without opening a bank account.³⁴ These transfers are explicitly labelled ‘remittances’ by the Central Bank of Russia. The figure shows a large drop in cross-border payments from Russia to Ukraine starting in 2014. Note that the drop before 2017 cannot be attributed to the mutual ban on foreign payment systems in Russia and Ukraine, which took effect only later in 2017. Thus, the drop before 2017 largely reflects the weakening economic ties and lower remittances from Russia to Ukraine as less Ukrainians worked in Russia (see Figure 10a).

Figure 10 suggests that the number of Ukrainians working in Russia and remitting money to Ukraine dropped sharply after the annexation of Crimea in 2014. Although these are national statistics and include Ukrainians not working in the border regions we think they also include many Ukrainians who previously worked in the Russian border regions and transferred their salary back to Ukraine via the banking system.

6.2. Empirical analysis

We now provide evidence for the effects of longer distances to open crossings on nighttime lights and plant exit. Since the border crossings were closed starting in March 2015, we exploit quarterly VIIRS nighttime lights data from 2012 onwards.³⁵ To this end, we estimate the following model:

34 This includes credit organizations, banks, money transfer systems and the Russian Post. Money transfer systems do not require to open a bank account but are included in the data.

35 Quarterly nighttime lights series are constructed from VIIRS data, which are not top-coded and better capture activity in low-lit areas. This is especially important for our study region along the northern border with Ukraine, where we have cross-border settlements in low-lit areas as well as large cities (e.g., Belgorod) located close to the border. The data are also available at a finer spatial resolution of 500×500 meters grid cells. We construct quarterly exit information for plants.

$$y_{it} = \beta_0 + \beta_1(\text{post}_{2015-Q1} \times \text{bigCity}_i) + \beta_2(\text{post}_{2015-Q1} \times \Delta \text{crossingsDist}_{it}) + \gamma_1(\text{post}_{2015-Q1} \times \Delta \text{crossingsDist}_{it} \times \text{bigCity}_i) + \alpha_i + \delta_t + \varepsilon_{it}, \quad (11)$$

where y_{it} is as before either the log of 1 plus the nighttime lights intensity of cell i or an exit dummy for plant p . In Section 6.2, $\text{post}_{2015-Q1}$ is a dummy variable taking value 1 starting in the first quarter of 2015 and zero otherwise; α_i are cell (or plant) fixed effects; and δ_t are year fixed effects, respectively. $\Delta \text{crossingsDist}_{it}$ is the log change in the shortest distance to the nearest open border crossing post-2015-Q1 compared to the shortest distance to the nearest open crossing before 2015-Q1. We measure the latter using either GC distance for plants and cells, or road ND for plants. We also use a weighted version of these measures to account for the fact that some border crossings are more important than others.³⁶ We also pay special attention to the distinction between big cities and rural regions and include a dummy bigCity_i . Indeed, the major international crossings are better connected to the large cities, which are thus affected differently. In particular, large cities generally saw much less change in their distance to the nearest open border crossing as the international crossing points all remained open.

We estimate Equation (11) for the set of (international and local) equipped border crossings in the four northern regions. We restrict ourselves to a shorter distance of either 50 or 100 km to focus on the effects going through the movement of people – which affects cross-border labour supply and cross-border shopping. As explained before, the local border crossings cannot be used for commercial merchandise trade. Thus, their closing does not materially change the geography of trade in goods.

Table 5 shows results for the relation between changes in distance to the nearest open border crossing and nighttime lights. We see that lights grew more strongly in large cities after 2015-Q1, but conditional on that less so in areas that experienced a substantial increase in distance from the nearest open border crossing. In other words, lights in the large cities that had better access to international border crossings that were not closed in the wake of the conflict grew more than lights in other places; and places where cross-border movements of people became more costly – as the distance to the nearest open border crossing increased – saw on average slower growth in lights. The major part of the economic cost of the border changes, as measured by nighttime lights, fell on rural areas that saw their distance to the closest open border crossings increase substantially.

Table 6 shows results with the exit dummy as the outcome variable. Columns (1)–(8) show that plants in big cities – essentially Belgorod at less than 50 km and Belgorod and Kursk at 100 km – saw more exit post-2015-Q1 compared to plants in more rural areas or smaller cities. Controlling for big cities, there are basically no effects of the increase in distance on plant exit: although almost all point estimates are negative, only one is weakly significant. Supplementary Appendix Table H.1 shows that the absence of

36 We did not find direct information on cross-border flows. We hence construct weights based on the NTL value of the cells of the crossing's main associated settlements (see Appendix A.3 for details).

Table 5. NTL and changes in distance to border crossings

	(1) Equipped points	(2) Equipped points	(3) All points	(4) All points
	GC	GCW	GC	GCW
Post-2015-Q1	0.080 ^a (0.000)	0.077 ^a (0.000)	0.082 ^a (0.000)	0.076 ^a (0.000)
Post-2015-Q1 × ΔcrossingDistance	−0.020 ^a (0.000)	−0.080 ^a (0.002)	−0.019 ^a (0.000)	−0.085 ^a (0.003)
Post-2015-Q1 × bigCity	0.364 ^a (0.014)	0.356 ^a (0.013)	0.393 ^a (0.024)	0.380 ^a (0.017)
Post-2015-Q1 × ΔcrossingDistance × bigCity	−2.978 ^a (0.606)	—	−1.466 ^b (0.580)	−5.381 ^b (2.171)
Cell fixed effects	✓	✓	✓	✓
Year-quarter fixed effects	✓	✓	✓	✓
Observations (cells×year-quarter)	8,216,500	8,216,500	8,216,500	8,216,500
R ²	0.875	0.875	0.875	0.875

Notes: The dependent variable is $\ln(\omega_i^{\text{VIIRS}} + 1)$, where ω_i^{VIIRS} is the luminosity of a 500 × 500 meters cell within a 50 km buffer from the northern part of the border, that is, Belgorod, Kursk, Bryansk and Voronezh regions. Columns (1) and (2) provide results for the equipped border points, while columns (3) and (4) provide results for all border points. Change in distance is measured by the GC distance to the nearest open border crossing. Standard errors are clustered at the cell level. The way we measure the distance change is indicated in the column header (GC = great circle distance (4); GCW = great circle distance weighted by border point settlements' NTL see (H.3)). The triple interaction coefficient in column (2) is not identifiable, because non-zero changes in distance (A.5) have systematically zero weight (H.2) associated with the closest point and vice versa (see Appendix A.3 for details). ^a p<0.01, ^b p<0.05, ^c p<0.1.

significant distance effects conditional on the big city dummy is probably driven by the fact that there is only little change in accessibility to the border for plants in large cities, whereas the bulk of the variation stems from plants in rural areas. As the major part of plants is clustered in and around the big cities, we have much less identifying variation than for cells, which may explain the much weaker results. Furthermore, exit is arguably more poorly measured using quarterly data than annual data, which introduces additional measurement error.³⁷ While this pattern in the data also affects the estimates in Table 5, the much larger sample size – especially for the rural areas – explains the much more precise estimates there. Overall, the triple difference specification seems very demanding on our data. We provide a number of additional robustness checks for our results in Supplementary Appendix D.3. We show, in particular, that the results are similar if we focus on a 100 km band along the border. This suggests that the largest part of the effect is due to what happens close to the border.³⁸

37 When exit is not directly reported to the authorities due to liquidation or accession, the Unified State Register of Legal Entities excludes companies that do not submit reports on taxes and levies or did not conduct banking transactions within the last 12 months.

38 We have also estimated the specification excluding the big city dummy. The qualitative results for nighttime lights do not change, but the coefficient of interest in the exit regressions becomes now negative and marginally significant.

Table 6. Plant exit and changes in distance to border crossings

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
		Equipped points				All points		
	GC	ND	GCW	NDW	GC	ND	GCW	NDW
Post-2015-Q1	0.045 ^a (0.005)	0.044 ^a (0.004)	0.045 ^a (0.004)	0.044 ^a (0.004)	0.044 ^a (0.005)	0.044 ^a (0.005)	0.044 ^a (0.004)	0.044 ^a (0.004)
Post-2015-Q1 x Δ crossingDistance	−0.006 (0.006)	−0.001 (0.007)	−0.029 ^c (0.016)	−0.021 (0.039)	−0.003 (0.006)	0.000 (0.006)	−0.026 (0.030)	−0.018 (0.040)
Post-2015-Q1 x bigCity	0.006 ^a (0.002)	0.008 ^a (0.002)	0.007 ^a (0.002)	0.007 ^a (0.002)	0.008 ^a (0.003)	0.008 ^a (0.002)	0.007 ^a (0.002)	0.007 ^a (0.002)
Post-2015-Q1 x Δ crossingDistance x bigCity	0.146 (0.158)	3.304 ^a (0.138)			−0.059 (0.099)	−0.018 (0.102)	−0.242 (0.490)	−0.056 (0.395)
Plant controls	✓	✓	✓	✓	✓	✓	✓	✓
Plant FE	✓	✓	✓	✓	✓	✓	✓	✓
Year-quarter FE	✓	✓	✓	✓	✓	✓	✓	✓
Observations (plants×year-quarter)	126,593	126,593	126,593	126,593	126,593	126,593	126,593	126,593
R ²	0.061	0.061	0.061	0.061	0.061	0.061	0.061	0.061

Notes: The dependent variable is a plant exit dummy ω_{pt} . The sample includes all plants that were active in 2015 within a 50 km buffer from the northern part of the border (Belgorod, Kursk, Bryansk and Voronezh regions). The 50 km buffer includes only Belgorod as a big city. Columns (1)–(4) provide results for equipped points. Columns (5)–(8) provide results for all points. Standard errors are clustered at the plant level. The way we measure distance is indicated in the column header (GC = great circle distance (4); ND = network distance (5); GCW = great circle distance weighted by border point settlements' NTL, (H.3); NDW = network distance weighted by border point settlements' NTL (H.3)). Standard errors are clustered at the plant level. The triple interaction coefficients in columns (3) and (4) are not identifiable, because non-zero changes in distance (A.5) have systematically zero weight (H.2) associated with the closest point and vice versa (see Appendix A.3 for details). ^a $p < 0.01$, ^b $p < 0.05$, ^c $p < 0.1$.

To summarize, we find some evidence that nighttime lights grew less in areas that experienced a larger increase in their distance to the nearest open border crossing compared to areas that did not experience such an increase. These effects are largely driven by the observations less than 50 km from the border and vary substantially across locations. Concerning plant exit, we only find more exit post-2015 in large cities but there are no strong spatial patterns. Overall, the empirical evidence for plant exit is not very strong, but we need to keep in mind that there is a systematic relation between the magnitude of the change in distance and city size since the international border crossings close to the larger cities remained open. This makes identification more complicated, especially for plants which are strongly concentrated in the large cities, and may explain why the results are mostly driven by big cities. Even if the exercise is demanding on our data, it suggests there are effects that may percolate through local cross-border labour movements.

7. CONCLUSION

Using the Russia–Ukraine conflict following the annexation of Crimea in 2014 as a natural experiment, we provide evidence that changes in borders – and the associated changes in market access – affect the economic outcomes of border regions substantially and sometimes quite locally. Exploiting the heterogeneity of the border changes between the south of Russia – which gained better market access to Crimea – and the north of Russia – which lost market access to Ukraine because of closed border crossings and tighter border controls – we find that regions more exposed to increased border frictions saw less growth in lights and more plant exit after 2014: compared to the least exposed regions, the most exposed regions saw 6–12% lower growth in GDP and a 43% increase in their manufacturing plant exit rate. Thus, the conflict inflicted substantial costs on some of the border regions.

7.1. Policy implications

While the direct economic consequences of conflict, due to, for example, destruction, have been studied for a long time, the indirect consequences due to changes in economic interactions are more elusive and harder to quantify. Our study adds to the recent literature that shows how conflicts generate contemporaneous spatial spillovers – by affecting non-combat areas – and how it may persistently affect conflict and non-conflict areas by disrupting cross-border relationships.

A first policy message is that the effects of conflict may significantly outlast the conflict itself, and that it may be impossible to move back to the pre-conflict regional equilibrium. One reason is the ‘legacy of war’ in the form of landmines and UXO, which reduce market access and give rise to human suffering and reduced economic performance in the long run. Both separatist groups in the Donetsk and Luhansk regions and Ukrainian government forces have extensively used landmines since the

conflict started in 2014, littering large swaths of eastern Ukraine: ‘Ukraine is severely contaminated with landmines and explosive remnants of war (ERW) from the armed conflict [...] Landmines have been documented in 11 of Ukraine’s 27 regions’.³⁹ This contamination is bound to affect market access for a long time. A second reason is the disruption of (cross-border) trade and production networks, as well as the erosion of trust. These mechanisms have been recently documented for Ukraine by Korovkin and Makarin (2023) and Korovkin *et al.* (2023a,b). We provide complementary evidence that suggests the conflict may also have disrupted cross-border employer–employee relationships and impeded cross-border shopping. The formation of relationships – trade, production, employment – entails sunk costs. These are likely to be large in the presence of specific physical capital and human capital investments, especially for the highly integrated border regions between Russia and Ukraine. Zhukov (2016, pp. 2 and 5) states that ‘the Donbas is home to numerous enterprises almost entirely dependent on export to Russia—such as locomotive plants and certain machine-building firms [...] Donbas machine-builders have been unable to find an alternative market for their goods. Their production is [...] highly specialized for the needs of Russian industrial clients’. In turn, the Russian industrial clients are likely highly dependent on material and personnel from Ukraine, which exacerbates the costs of severing relationships. Given that switching trading partners, or reorganizing production networks, or switching employers are both difficult and costly, the conflict-induced changes could be permanent, even if we were to revert to the pre-conflict status quo and if trust were restored. In other words, if there are multiple equilibria and if sunk costs are important, conflict can shift the economy permanently down a new – potentially inferior – equilibrium path. Finally, a third persistent ‘legacy of war’ is due to investments in durable infrastructure such as the power grid and the bridge to Crimea. These large sunk investments are likely to induce path-dependence and persistence, making it again unlikely that we would revert back to a pre-conflict equilibrium. Conflict is thus likely to shape the economic path for a long time to come.

The second important take away from our analysis for policy is that the annexation of territory is unlikely to yield large net economic gains.⁴⁰ Conflict begets conflict, and countries are likely to get stuck in a ‘conflict trap’ as they start to retaliate and escalate hostilities (Rohner *et al.*, 2013). One may thus ask the deeper question whether a country can really ‘gain market access’ via annexation? Of course, the annexed territory will – by definition – be easier to access, but retaliation, either from the international community, or from the country of the annexed region, or from both, may offset any gains. Taking away a ‘small piece’ may lead to the loss of the ‘larger part’. We have documented the differential regional effects on the country that annexes another’s territory.

39 Source: https://www.hrw.org/news/2023/06/13/landmine-use-ukraine#_ftnref1

40 Of course, throughout history, annexations have involved numerous non-economic reasons such as military considerations, shared history, and cross-border ethnic ties. These non-economic reasons are often much more important than economic ones.

Although our DiD approach, which washes out country-wide effects such as international sanctions, does not allow us to determine whether there are regional winners in absolute terms, there could be regional losers that suffer from the combination of changes in market access, due to retaliation, and international sanctions following the annexation.

SUPPLEMENTARY DATA

Supplementary data are available at *Economic Policy* online.

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APPENDIX MATERIAL

This set of appendices is structured as follows. In [Appendix A.1](#), we provide a brief description of our nighttime lights data. In [Appendix A.2](#), we explain our plant-level data. Last, [Appendix A.3](#) provides details on the other variables and measures used in the study. Additional details are provided in an extensive separate set of Supplementary appendices.

APPENDIX A: DATA

Appendix A.1: Nighttime lights

Datasets. We rely on publicly available nighttime lights satellite data based on the DMSP Operational Linescan System (DMSP, for short) and Suomi NPP VIIRS (VIIRS, for short). DMSP and VIIRS were developed for different purposes and thus measure lights differently. Whereas the former has a relative coarse measure between 0 (no lights) and 63 (most intensive lights, top-coded), the latter measures lights more continuously, better captures low-lit areas, and is not top-coded. DMSP was discontinued in late 2013 and gradually replaced by VIIRS starting 2012. The former offers a spatial resolution of 1×1 kilometre grid cells, whereas the latter provides a resolution of 500×500 m.

To our knowledge, [Li et al. \(2020\)](#) are the first to harmonized the DMSP and VIIRS series globally. The harmonized NTL series (HNTL, for short) comprises temporally calibrated DMSP nighttime lights (1992–2013) and DMSP-like nighttime

lights from VIIRS (2014–2018) using 2013 to assess the relationship between DMSP and VIIRS. The HNTL series of [Li et al. \(2020\)](#) spans 1992–18 at a spatial resolution of 1×1 km cells. We focus on the period 2005–18. The lights intensity is measured by a number ranging from 0 to 63.

The VIIRS data cover a more recent period and go back a little before 2014. We utilize quarterly VIIRS nighttime lights from 2012-Q2 to 2018-Q4 to estimate changes in NTL intensity in response to changes in distance to the nearest border crossing in Section 6 (see [Table 5](#)). There, we also use the VIIRS data to compute the intensity of lights at night for the settlements associated with cross-border movements (see [Appendix A.3](#)). A more detailed description of our nighttime lights data, its preparation and post-processing, are relegated to [Supplementary Appendix B.1](#).

Cell-level GDP apportioned using lights. To approximate GDP at the cell level, we combine the HNTL data with regional GDP data from Goskomstat for Russia and from Ukrstat for Ukraine. Formally we construct a lights-weighted measure of GDP for cell i in region r , denoted $\omega_{i(r)t}$, in year t as follows:

$$\omega_{i(r)t} = \text{GDP}_{rt} \frac{\text{NTL}_{it}}{\sum_{k \in r} \text{NTL}_{kt}}, \quad (A.1)$$

where NTL_{it} denotes cell i 's raw NTL luminosity. [Equation \(A.1\)](#) simply apportions regional GDP to each cell in the region based on its share in regional luminosity. While this procedure obviously has limitations, we think it at least partly allows us to control for large differences across regions (due to, e.g., regional specialization in high vs. low value-added activity or more urban vs. more rural regions).

We use [Equation \(A.1\)](#) to construct a GDP-based measure of market potential for Russian border regions and plants. The counterpart of [Equation \(2\)](#) based on GDP instead of raw lights is given by

$$\text{GMP}_{it}^{\text{UKR}} = \sum_{r \in \mathcal{R}_{\text{UKR}}} \sum_{j \in r} \omega_{j(r)t} e^{-\alpha d_{ij}}, \quad \ln(\exp_i^{\text{GDP}}) = \ln\left(\frac{1}{5} \sum_{t=2009}^{2013} \text{GMP}_{it}\right), \quad (A.2)$$

where $\mathcal{R}_{\text{UKR}}$ denotes the set of regions in Ukraine.

How are our regressions using raw lights and lights-weighted GDP related? Our two regressions – using either raw NTL ω_{it} or the GDP-weighted measures $\omega_{i(r)t}$ – are as follows: $\ln \omega_{it} \propto \gamma_2 \times \text{post}_{2014} \times \text{lnexposure}$ and $\ln \omega_{i(r)t} \propto \gamma_2 \times \text{post}_{2014} \times \text{lnexposure}$. Since $\ln \omega_{i(r)t} = \ln \omega_{it} + \ln \text{GDP}_{r(i)} - \ln \sum_{j \in r} \text{NTL}_{jt}$, the latter can be rewritten as follows:

$$\ln \omega_{it} + \ln GDP_{r(i)t} = \gamma_2 \times post_{2014} \times \ln exposure + \ln \underbrace{\sum_{j \in r} NTL_j}_{\xi_{rt}}, \quad (A.3)$$

where ξ_{rt} is a term that is absorbed by region–year fixed effects. The foregoing expression shows that we need to adjust the magnitude of our NTL estimates by the GDP term to make the magnitudes of the two estimates comparable. Let ϵ denote the elasticity of GDP per capita with respect to average regional NTL: $\ln GDP_{rt} = \epsilon \times \ln \bar{\omega}_{rt} - \ln POP_{rt}$, so that Equation (A.3) can be rewritten as follows:

$$\ln \omega_{it} + \epsilon \times \ln \bar{\omega}_{rt} = \gamma_2 \times post_{2014} \times \ln exposure + \xi_{rt}, \quad (A.4)$$

where $\ln POP_{rt}$ is subsumed in ξ_{rt} . Thus, to make the magnitudes of the estimates in the raw NTL and in the GDP-weighted NTL regressions roughly comparable, we need to divide γ_2 by $1 + \epsilon$ to control for the fact that lights-weighted GDP varies more since GDP itself changes with lights.

Appendix A.2: Plant-level data

Data sources. Our plant-level data come from two main providers: Interfax’s SPARK and Bureau van Dijk’s Ruslana. We identify individual plants using national identifiers – Russian National Nomenclature of Businesses and Organizations (OKPO) and Tax Identification Number (INN). We geo-reference plants using the Yandex Maps API service to map the de facto address – where a plant operates – to geographic coordinates. Each plant reports a date of entry, a date of exit, and its primary activity according to the Russian Classification of Economic activities (OKVED). The classification changed in 2014. For some plants registered after 2014, we only know their OKVED2014 code. We thus create a concordance between OKVED2007 and OKVED2014. Additional details on plant-level data and their treatment are provided in [Supplementary Appendix B.2](#).

Plant Exit. We construct the exit reason for all plants within 300 km distance from the border using the Ruslana and SPARK Interfax databases. We have tried to make these statuses as comparable as possible across the two datasets.⁴¹ There are many reasons for plant exit. The top-5 exit statuses and their shares in the sample are reported in [Appendix B.2](#). The first two reasons (about 50% of the cases) relate to firms that have simply ceased any activity – no banking or fiscal operations are

41 Unfortunately, this is not perfect for several reasons: (1) Ruslana and SPARK have semantically similar statuses from the Unified State Register of Legal Entities, but in some cases using a different wording or formulation; (2) in the latest updates we extracted in 2021 we have detailed statuses in Russian, whereas they were in English in our previous extracts; and (3) SPARK provides their own aggregate classification that we partly used in our previous analysis and which we had to unfold again.

carried out anymore. There has, however, not been a formal bankruptcy or liquidation procedure for these firms. These include mostly small firms that disappear and go out of business. The next two reasons (about 30% of the cases) relate to formal bankruptcy or liquidation procedures involving the courts. These are usually larger firms where many stakeholders try to secure claims during a formal liquidation of the firm. The last reason corresponds to firms that do not maintain an administrative record in compliance with tax requirements, that is, the information about either the location or the owner is not ‘trustworthy’. Such firms must update their information within 6 months or else they are considered liquidated and removed from the State Register. Thus, close to 90% of what we consider ‘plant exit’ is administratively related to: (1) no more activity or (2) a formal liquidation or bankruptcy procedure. Note that exit may also occur through accession – whereby a new party acquires a majority stake in the company – or mergers. These two categories are also present in our data, but they account for a small part of the cases of exit. Thus, we think that our measure of ‘exit’ captures ‘true exit’ in most cases.

Appendix A.3: Other variables and measures

Distance measures. We measure – for each NTL cell and plant – its distance to the positive border segment in the south and the negative border segment in the north. We compute the GC distance to the closest border vertex using either the plant’s coordinates or the NTL cell’s centroid. For each plant, we also use OpenStreetMap’s road network layers to compute the shortest path using the ND to the nearest border customs points. For plants in the south, access to Crimea is via the sea of Azov or the Black Sea.⁴² We construct the shortest path to the sea ports providing access to Crimea as the ND to the positive border. Additional details are in [Supplementary Appendix G](#).

Geographic variables. We first construct various distance band dummies, either for the border in general or for the positive and the negative border segments separately.⁴³ Second, we use settlement polygons and points from OpenStreetMap to construct a ‘big city’ dummy that takes value 1 if the observation is located in a city with at least 300,000 inhabitants. Third, we create a categorized variable indicating whether the plant or the NTL cell lies to the south ($\text{lat} < 47.14$), in the same latitude ($47.14 < \text{lat} < 49.89$), or north ($\text{lat} > 49.89$) of the Donbass. Last, we collect data on the precise locations of all international border crossings between Russia and

42 We disregard the recent bridge across the strait of Kerch that was only opened to traffic in May 2018.

43 We select 150 km as our preferred specification because it captures most of the big cities along the border. In robustness checks, we also use 50 and 100 km. The 100 km band cuts through many cities, whereas the 50 km band has a substantially smaller sample size with much less economic activity.

Ukraine; and we construct data for all local border crossings that exist to simplify local cross-border movements between bordering regions in Russia and Ukraine. We provide details in [Supplementary Appendix H](#).

Measure of exposure to sanctions. First, we compute the total value of exports to Russia from countries imposing sanctions in the sanctioned and embargoed products. The former are extraction equipment and double-use products, whereas the latter are mainly food products. Next, we construct a crosswalk from the HS product classification to OKVED2007 industry codes (see the intermediate steps in [Supplementary Appendix B.2](#) for details). This allows us to construct a raw measure of the share of plants in industries under sanctions in the total number of plants in each municipal district, which we average over 3 years before the conflict. Next, we compute per plant trade values in each sanctioned industry and municipality. Last, we sum these values and weight them by the share of sanctioned plants in the municipality. Since our data do not allow us to do anything more detailed, we assume that plants in non-sanctioned industries had zero trade in sanctioned goods.

Other variables. We obtain yearly data on municipal populations in Russia between 2012 and 2018 from Goskomstat's Database of Municipal Districts. Because those series fluctuate too much to be reliable in their time dimension, we use their averages. We further obtain GDP for each region from Goskomstat for Russia and from Ukrstat for Ukraine. Regional GDP for Russia is provided for the years 2004–19 in current prices. Since Crimea and Sevastopol became subjects of the Russian Federation in 2014, we construct GDP series for these regions pre-2014 using official statistics from Ukrstat in current US dollars. We use the average exchange rate between US dollars and Russian rubles pre-2014 from the IMF to compute GDP values for Crimea and Sevastopol.

Changes in the distance to the nearest open border crossing. We measure changes in distance to the nearest border crossing point as follows:

$$\Delta_{\text{crossingDistance}}^D_p = \ln \left(\frac{\min_{j \in \mathcal{B}_{\text{post-2015}}} d_{p,j}^D}{\min_{j \in \mathcal{B}_{\text{pre-2015}}} d_{p,k}^D} \right), \quad (A.5)$$

where $d_{p,j}^D$ is distance measured as great-circle or NDs $D \in \{GC, ND\}$ between plant p and border crossing point j . The set $\mathcal{B}$ is defined as either all border crossing points or the set of equipped border crossing points only.

Since we do not observe statistics on the volume of cross-border movements before and after the conflict, we use the intensity of nighttime lights as a proxy for the importance of each border crossing point. We proceed in several steps. First, for each settlement associated with the border crossing point on the Russian and the

Ukrainian sides, we compute the sum of 500-by-500 m cell luminosity from the annual VIIRS dataset.⁴⁴

Next, we compute the average intensity of nighttime lights for each settlement for the years 2013–2014. We also compute the geodesic distance between centroids of settlement pairs. With this at hand, we build weights for each border crossing point using a gravity-like relation: the weight is the product of the average pre-conflict light intensity of a settlement pair divided by the squared distance between them.

Finally, we normalize these weights to fall into the interval $[0, 1]$ and use them either as analytical weights or as multipliers for changes in distance to the border crossing points in our regressions. We only report the latter results, but the former are qualitatively identical and available upon request. A more formal exposition of the above steps is relegated to [Supplementary Appendix H](#).

⁴⁴ Settlement polygons are gathered from the OpenStreetMap database. If a settlement is defined as a point instead of a polygon, we put a buffer around it with a radius equal to the average distance between border crossing settlement's vertices in Russia and Ukraine, respectively.